A Plan for Refactoring the pmsimstats-ng Whitepapers into One Series

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Purpose. This document reviews the whole of `docs/` (58 numbered whitepapers, 5 planning files, 6 other files) for consistency of notation, methods and conclusions, and for redundancy, and sets out a plan for rebuilding the set as one series that can be read from start to finish. The series is meant to carry more detail than the manuscripts under `analysis/report/`: every derivation, every table and every number, each traceable to a script and an output file. The plan states the design principles of the series, its Parts and chapters, where every current file goes, the conflicts to settle before content is moved, the gaps to fill, and the order of work. It incorporates the state of the simulation work on 2026-10-11 and three user decisions of the same day (Section 2.11) and the ten decisions recorded in Section 10.

Contents

Notation and evidence . . . 4
1. Findings summary . . . 4
1.1 State of the corpus . . . 4
1.2 Classes of problem . . . 5
2. Design principles for the refactored series . . . 6
2.1 Audience and purpose . . . 6
2.2 Relation to the manuscripts . . . 6
2.3 Relation to the mathematics compendium . . . 7
2.4 Relation to NOTATION.md . . . 7
2.5 One evidence-label vocabulary . . . 7
2.6 One Monte Carlo methodology . . . 8
2.7 One analysis-ID scheme . . . 8
2.8 Canonical names . . . 9
2.9 Superseded material: the dated-note policy . . . 10
2.10 Chapter template . . . 10
2.11 User decisions of 2026-10-11 . . . 10
3. Target structure . . . 11
3.1 Front matter . . . 11
3.2 Part I. The problem and the published process . . . 11
3.3 Part II. The data-generating process . . . 12
3.4 Part III. Theory of the interaction test . . . 15
3.5 Part IV. Analysis of the interaction . . . 15
3.6 Part V. The treatment main effect and application . . . 17
3.7 Part VI. Notes for the original authors . . . 17
3.8 Appendices and Volume II . . . 18
4. Disposition of every current file . . . 20
5. Conflict register . . . 24
5.1 Resolved by later work: needs propagation only . . . 24
5.2 Genuinely open: needs analysis, inspection or a decision . . . 28
5.3 Needs the pending reruns, or the reporting of finished ones . . . 32
6. Redundancy map . . . 33
7. Notation and convention fixes . . . 36
7.1 Documents before 34: wholesale . . . 36
7.2 Regressions in 34 to 58 . . . 37
7.3 Cross-cutting rules for the series . . . 38

8. Gaps . . . 38
9. Migration plan . . . 42
9.1 Phases . . . 42
9.2 File layout and naming . . . 43
9.3 How old numbers keep resolving . . . 43
9.4 Updating citations in manuscripts, CM and scripts . . . 44
9.5 Keeping the project `CLAUDE.md` in sync . . . 44
10. Decisions . . . 45
11. Evidence basis and limits of this plan . . . 45
References . . . 46

List of Tables

1 Evidence labels used in this plan . . . 4
2 Numbered documents by status (survey; classification merged across the four surveys) . . . 4
3 Numbered documents by format (observed, `ls`; review status from the surveys) . . . 5
4 The evidence-label vocabulary of the series (merging the sets of `docs/51`, `docs/53`, `docs/54`, `docs/55`, `docs/56`, `docs/57` and `docs/58`) . . . 7
5 The analysis-ID scheme proposed for the series . . . 8
6 Front matter of the series . . . 11
7 Part I of the series . . . 12
8 Part II of the series . . . 13
9 Part III of the series . . . 15
10 Part IV of the series . . . 15
11 Part V of the series . . . 17
12 Part VI of the series . . . 18
13 Appendices of the series . . . 18
14 Outline of Appendix I, the legacy record (decided 2026-10-11) . . . 19
15 Disposition of every file in `docs/` (survey recommendations; disagreements resolved in the rows for 09a, 19 and 32) . . . 20
16 Conflicts resolved by later work (propagation only) . . . 25
17 Conflicts that need analysis, inspection or a decision . . . 29
18 Conflicts that wait on reruns or on reporting finished runs . . . 33
19 Redundancy clusters and their single home . . . 33
20 Notation and convention fixes for content carried from documents before docs/34 . . . 36
21 Notation regressions in docs/34 to docs/58 . . . 37
22 Gaps the series must fill . . . 39
23 Migration phases (computational gaps GP-15 to GP-22, GP-25, GP-31 to GP-33 and GP-39 run alongside, on the server) . . . 42

Notation and evidence

Notation follows `analysis/report/NOTATION.md` (revision 2026-10-09). This plan introduces no new mathematical symbol. It uses four kinds of local label, none of which is a symbol: chapter identifiers `ch01` to `ch28`, front-matter identifiers `fm0` to `fm2` and appendix letters `A` to `J` (Section 3); conflict identifiers CR-01 onward (Section 5); gap identifiers GP-01 onward (Section 8); and decision identifiers DQ-1 onward (Section 10). Old documents are cited as `docs/NN`, with section numbers as they stand on 2026-10-11.

Table 1: Evidence labels used in this plan

Label	Meaning in this plan
observed	read in the source document, a file listing or a command output on 2026-10-11 by the author of this plan
spot-checked	a survey claim re-read in the source at the cited section or line on 2026-10-11
survey	reported by one of the four survey files listed below and not re-checked here
inferred	a judgment drawn from the pattern of what was read, not checked
decided	a user decision of 2026-10-11 (Sections 2.11 and 10)

The evidence base is four survey files written on 2026-10-11 that between them read every file in `docs/` in full: survey a (docs/00 to docs/20, including the `.tex` set, 09a and 19), survey b (docs/21 to docs/33 and the planning files), survey c (docs/34 to docs/45) and survey d (docs/46 to docs/58, the guides, the `pub_review` files and the slide deck). Read directly for this plan: `NOTATION.md` in full; the `\part`, `\section` and `\subsection` outline of `analysis/report/compendium-mathematics.tex` (written CM below; 8,723 lines, 4 parts, 23 sections); the YAML titles of the 14 manuscripts; the headings of the paper 04 working protocol `analysis/scripts/treatment-main-effect/00-ademp-protocol.md` (written P04 below); the project `CLAUDE.md`; `docs/57` Sections 1 and 8.1 to 8.2; `RERUN-anchored.md` Sections 1, 2 and 7; and the summary files of the anchored reruns. Section 11 lists what was not checked.

1. Findings summary

1.1 State of the corpus

The top level of `docs/` holds 116 files besides this plan: 58 numbered documents, 5 planning files, 6 other files, and their PDFs (observed, `find`). Four subfolders (`archive/`, `figures/`, `references/`, `reviews/`) and a `.claude/` settings folder sit beside them.

Table 2: Numbered documents by status (survey; classification merged across the four surveys)

Status (as judged by the surveys)	Count	Documents
Current within its stated scope	13	22 (parity workflow), 26, 35, 36, 37, 39, 47, 51, 53, 55, 56, 57, 58
Partly superseded	16	04, 06, 08, 19, 24, 25, 31, 32, 34, 40, 41, 42, 45, 48, 49, 54
Superseded or historical record	22	00, 01, 02, 03, 07, 07a, 09, 09a, 13, 14, 15, 16, 17, 18, 20, 21, 23, 28, 33, 46, 50, 52
Planning	4	22 (journal targets), 27, 38, 44
Reference	3	29, 30, 43
Total numbered	58	numbers 05, 10, 11, 12 are absent; 22 is used twice; 07a and 09a are lettered

Table 3: Numbered documents by format (observed, `ls`; review status from the surveys)

Format	Count	Notes
`.tex` only	16	00, 01, 03, 04, 06, 07, 07a, 08, 09, 13, 14, 15, 16, 17, 18, 20; none reviewed on 2026-10-07 or swept by the 2026-10-09 notation pass; only 07a has a PDF, and it predates the repository (December 2025)
`.md` and `.tex`	2	09a and 19: the `.md` was reviewed and corrected, the `.tex` was not, and the two now disagree
`.md`	39	21 to 58; each has a PDF at least as new as its source (survey, mtimes)
PDF only	1	02, an annotated review copy of a deleted `.tex`

Four facts about the simulation work frame the plan (observed on disk, 2026-10-11, and from the coordinator's brief):

- The anchored stress-test main run (`simrun` job 15, server) has finished: `summary-full-anchored.csv` holds 451 rows, the 41 cells of `docs/46` times its 11 original analyses (S, C1, C2, A, ES, EA, F1, F2, R, DS, DA). By `RERUN-anchored.md` Section 1 the driver's default is anchored carryover with randomized path allocation, so this is the stress test under the refined process (inspected). No document reports it.
- The local anchored `--extra` (S, G1, G2) and `--extra2` (S, F1A) arms are not paired with job 15 (different BLAS and LAPACK draws; 0 of 28,000 trials match). They are being rerun on mercury as jobs 21 and 22 to pair with job 15 (decided).
- The local anchored main pass of the covariance study has finished (`summary-alt250-null500-anchored.csv`, 320 rows; fixed allocation, per `RERUN-anchored.md` Section 1). Its follow-up and $D_{bc,it}$ passes were cancelled. No document reports the anchored main pass.
- The data-chosen half-life extension (job 14; 58 cells, 40,000 trials, refined process) is reported in `docs/57` Section 8, with F1A0B as the provisional data-driven primary (spot-checked, `docs/57` Section 1, lines 209 to 223).

1.2 Classes of problem

Notation. The 16 `.tex` documents and the `.tex` halves of 09a and 19 predate every notation decision. They use the retired labels "Architecture A" and "Architecture B", "N-of-1" for the Hybrid design, "ER" and "TR" for PB and TV, $S_{i,t}$ or `Sx` for Y_{it}, ρ for the compound-symmetry correlation (canonical ρ_{cs}), c_{bm} for the mean-moderation multiplier (canonical β_{bm}, rule 2), and D_{bc} for the binary indicator (rule 1) (survey a). The documents 34 to 58, swept on 2026-10-09, are clean in substance but carry regressions: reserved letters reused (q in 35 and 36, f in 38, 39, 41, 42, 45, Δ in 35, M, m, g, h, ω in 57, B, G, K, Z in 58), plain $E(\cdot)$ in 37, $N(\cdot)$ in 41, one letter with several meanings across documents (v, ω, C, μ), and one object with several symbols ($\tilde{r}$, r, $\tilde{g}$ for the padded coupling vector; n_{on} against n_2 for row counts; σ_v^2 against τ^2) (surveys c and d). Section 7 lists them.

Methods described differently. The same quantity is quoted at three parameter points without saying which: the Hybrid ceiling is 0.456 (`docs/34`, configuration F, $\rho = 0.7$), 0.291 (`docs/36`, configuration F, $\rho = 0.8$), 0.440 (`docs/36`, configuration C, $\rho = 0.8$) and 0.480 (`docs/35`, configuration C, $\rho = 0.7$), each correct for its construct (survey c). The stress-test and covariance-study numbers in 45 to 56 rest on the recursive carryover and fixed allocation of the stored runs, while 57 Section 8 and job 15 use the refined process (survey d; observed). At least three incompatible lists of differences between the package and the published process exist (`docs/23` Section 2, `docs/32` Section 2, `docs/54` Section 1), and NOTATION Part 3 gives a fourth (survey b). Size bands differ (0.043 to 0.057 over 8 cells in 45; 0.0458 to 0.0542 in 43; 0.046 to 0.054 over 12 cells in 57), and evidence-label vocabularies differ in at least six ways across 46 to 58 (surveys c and d).

Conclusion conflicts. The most consequential, all detailed in Section 5:

1. The recommended interaction analysis is stated five ways: the plain `corCAR1` model (24, 26, 31, `CLAUDE.md`), `us-KR` with $D_{bc,it}$ (45), F1 at a fixed half-life (49 Section 7, 50 Section 2, 56), and F1A0B as data-driven primary with F1 as fallback (57 Section 1, revised 2026-10-10).
2. The mechanism of the published carryover collapse: standard-error inflation with an unbiased estimate (03, 04, 09, 02) against attenuation because the coupling gap is zero (36 Section 5.3, 37 Section 5.1, 55 Section 5).
3. The size of the architecture contrast: 10 to 15% against 40 to 60% relative loss (02, 19, `CLAUDE.md`) against 6.9% against 15.4% at matched total $N = 70$ (NOTATION Part 2).
4. Coupling ceilings: "0.25 to 0.49+", "about 0.6", "about 0.45" (03, 07a, 19.tex, `CLAUDE.md`) against 0.256 under compound symmetry and 0.474 (0.480 in Hybrid) under the separable form at $\rho = 0.7$ (34, 35, 52 Section 2).
5. The cost of F1 relative to S: 29% at the reference cell of the stored run (47 Section 1, line 123, spot-checked: S 0.514, F1 0.364), never re-estimated under the refined process, where F1 alone has 0.438 (57 Section 8.3, line 1544, spot-checked). Job 15 now holds both.

Redundancy. Thirty-three clusters of overlap were found across the four surveys (Section 6). The worst are: `docs/24` against `docs/31` Parts I to III (near verbatim, each holding corrections the other lacks); `docs/40` Section 9 against `docs/42` Sections 2 and 4 (identical tables and figures); `docs/46` Amendment 2 against `docs/47`; the "superseded by F1" paragraph repeated in six documents (35, 41, 42, 43, 44, 45); the description of the published process in nine places (24, 25, 28, 31, 36, 37, 40, 44, 53); the ceiling formula and tables in eight (32, 34, 35, 36, 37, 44, 54, CM); package implementation status maintained in at least five (34, 35, 36, 44, 52), each with its own inspection date; and `docs/23`, which copies 21 earlier documents.

Convention lapses. Every `.tex` document has a hand-written "Rendered on" footer with the obsolete path `~/prj/alz/10-pmsimstats-ng` and no notation index, Purpose or evidence labels (survey a). Further lapses: "Working draft" in the header of `docs/36`, the most-cited document (cited by 20 others); a word the user's style rules forbid in `plan-coherent-analysis-comparison.md` line 20 and in the `pub_review` files; British spellings in `reorg-plan-2026-04-22.md` and in `CLAUDE.md`; "patient(s)" in 37, 38 and 39 against "participant"; "hybrid" in lower case in 28 and 31; the file name of `docs/51` ("ten") against its seventeen features; two numbers 22; swapped figure files in `docs/40`; one orphan figure (`36-fig4-contrast-reps100.png`, cited nowhere) (surveys a to d).

Stale PDFs and `.tex`. No current PDF exists for any of the 16 `.tex`-only documents; 07a's PDF is from 2025; dated duplicate PDFs of 29 and 30 from June remain; `docs/18` cannot render (its figures under `analysis/figure5/` are gone); `docs/02` is a PDF of a deleted source (survey a; survey b; observed).

2. Design principles for the refactored series

2.1 Audience and purpose

The series is written for a methodologist who has to understand, reproduce or extend the work: a statistician joining the project, a referee of a manuscript, or one of the original authors. It assumes graduate statistics and no prior knowledge of the project. It is the long form of the program. Where a manuscript states a result and cites its derivation, the series gives the derivation steps, the construct, every cell of every table, the Monte Carlo error and the command that regenerates the number. It is not a manuscript: it records the history of decisions in one appendix rather than throughout, and it can be longer than any journal allows.

Three reading paths are offered in the reading guide (`fm0`): the full path (Parts I to VI); the practitioner's path (`ch01`, `ch02`, `ch10`, `ch22`, `ch23`, `ch25`); and the original authors' path (`ch03`, `ch04`, `ch26` to `ch28`).

2.2 Relation to the manuscripts

Each chapter names the manuscripts that use it and the manuscript sections that depend on it. The rule is one direction of detail: a manuscript may summarize a chapter, and a chapter never depends on a manuscript

for a number it does not itself tabulate. Every number in a chapter carries a source line (script, output file, commit) in its table caption or in the chapter's "Reproduce" section. Where a manuscript reports a result the series reproduces, the chapter states whether the numbers agree and, if not, why (for example the published power at $t_{1/2} = 0.1$ week, quoted as 0.126, 0.078, 0.100 and 0.076 in 37, 53, 55 and paper 01; reconciled in 55 Section 7.3, survey d).

2.3 Relation to the mathematics compendium

Three placements were considered.

1. **The compendium as the series' mathematical volume (decided 2026-10-11, DQ-1).** CM stays one LaTeX document and becomes Volume II of the series. Chapters state each result, give the derivation in enough steps to follow the argument, and cite the CM subsection label for the full proof and its verification script. Derivations that exist only in a whitepaper (57 Sections 3 to 4, 58 Propositions 1 to 7, 55 Section
   5) are moved into CM once, and the chapter keeps the argued form.
2. **A separate reference.** CM stays as it is, cited ad hoc. This is the status quo; it leaves the reader without a path through CM and lets the two drift (CM's main-effect section still describes version 1 of paper 04, and CM cites none of 55 to 58; survey d).
3. **Folded into the chapters.** Each CM section moves into the chapter it serves. This gives one document per topic but destroys CM's LaTeX structure, its equation labels and its pairing with the `analysis/scripts/compendium-mathematics/check-*.R` verification scripts, and it is the largest and riskiest single move of the whole refactor.

Option 1 was recommended, and adopted, because the main defect the surveys found is redundancy, and two full copies of each derivation would recreate it; because CM already has a reading order (its Section "Scope and reading order") and verification scripts that a Markdown chapter cannot hold; and because 46 of the 118 files that cite `docs/NN` outside `docs/` are under `analysis/report/` and already cite CM labels (observed, grep). The chapter template (Section 2.10) gives each chapter a "Derivations" section with a row per result: statement, CM label, verification script. The cost of option 1 is that CM must be brought up to date (GP-28) and its Part order aligned with the series Parts (Section 3 shows that the existing order already matches closely).

2.4 Relation to NOTATION.md

`analysis/report/NOTATION.md` stays the single source for symbols, code identifiers and named processes. The series adds nothing to it without a NOTATION revision. Each chapter's notation block lists only its local symbols, with a status column (canonical or local), and links every canonical symbol to NOTATION by its table name. A local symbol may not use a letter on NOTATION's reserved list. The front-matter chapter `fm2` holds the project vocabulary that NOTATION Part 3 holds for the manuscripts (named processes, configurations, analysis IDs, evidence labels), so that the lexicon (Appendix E) can stay a cross-field glossary.

2.5 One evidence-label vocabulary

The series uses seven labels, defined once in `fm2` and used in every chapter:

Table 4: The evidence-label vocabulary of the series (merging the sets of `docs/51`, `docs/53`, `docs/54`, `docs/55`, `docs/56`, `docs/57` and `docs/58`)

Label	Use
verified	computed or run, and the output observed; cite the script
derived	proved analytically; cite the CM label or the chapter section

Label	Use
inspected	read in code or in a source document; cite file and line or commit
quoted	taken from another document; cite document and section
argued	a reasoned position, not a proof
inferred	consistent with the surrounding pattern, not checked
unverified	not checked at all

2.6 One Monte Carlo methodology

`fm2` states, and every chapter applies, the following conventions, collected from 45 Section 2, 46 Section 3, 57 Section 8.1 and CM "Shared performance measures":

- **Reporting frame.** ADEMP for every study (aims, data-generating mechanisms, estimands, methods, performance measures), with the construct box of Section 2.10.
- **Size band.** For a pooled null rate over n null trials the band is $0.05 \pm 2\sqrt{0.05 \times 0.95/n}$, stated in the caption with n; for 12,000 trials, 0.046 to 0.054. Per-cell cap 0.075; a prognostic cell above 0.075 raises a flag. These are the screens of `docs/46` as amended (quoted, 57 Section 8.1).
- **Distances from nominal.** "k MCSE above 0.05" uses the MCSE at 0.05, $\sqrt{0.05 \times 0.95/n_{\text{reps}}}$, as `docs/42` Section 6 does; the convention of `docs/40` Section 9 (MCSE at the observed rate) is retired (resolves CR-26).
- **MCSE with every number.** Every rejection rate carries its binomial MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$; every pooled rate its pooled MCSE; every difference between independent runs $\sqrt{\text{MCSE}_1^2 + \text{MCSE}_2^2}$ (`RERUN-anchored.md` Section 7); paired comparisons within a run use McNemar.
- **Warp-speed bootstrap.** Rates from one resample per trial (Giacomini, Politis and White, 2013) are labeled warp-speed; their MCSE is computed by resampling trials and is larger than the binomial one (0.0097 to 0.0112 against 0.0069 per cell at $n_{\text{sim}} = 1{,}000$; quoted, 57 Section 8.1). The binomial MCSE is never quoted alone for a warp-speed rate.
- **Size-adjusted power** follows CM "Size-adjusted power"; maximin shortfall states its cell set in the caption (non-prognostic alternative cells, or all 26 alternative cells), so the two readings of `docs/43` and `docs/47` Section 7 stay apart (CR-29).
- **Replicate bands.** Null cells $n_{\text{reps}} \geq 1{,}000$ and alternative cells ≥ 500 for any number that enters a decision; smaller runs are labeled "illustration" or "pilot" in the caption.
- **Sample size.** N is always the total (NOTATION rule 4), and every study table states the allocation (fixed or randomized).

2.7 One analysis-ID scheme

The stored interaction-track labels (S, C1, C2, A, ES, EA, F1, F2, R, DS, DA, G1, G2, F1A, F1A0, F1A0B, F1AB, MA, MaxT and their suffixes) are keys in the stored stress-test files, and NOTATION Part 2 forbids a display-time mapping layer. The scheme (decided 2026-10-11, DQ-5) therefore keeps them and changes what is new or local. A document names the track at first use ("interaction analysis F1", "main-effect analysis `M-F1`").

Table 5: The analysis-ID scheme proposed for the series

Class	Form	Examples	Change
Interaction track (stress test, covariance study selection, data-chosen half-life)	bare label	S, F1, F1A0B, G2	none; the default track of Parts III and IV

Class	Form	Examples	Change
Main-effect track (P04, `docs/58`)	`M-` prefix	`M-S`, `M-F1`, `M-VF`, `M-RV`, `M-F1A`	rename in P04, `docs/58` content and `ch24` before paper 04's stored data exist (DQ-5, decided)
Covariance study analyses (`docs/45`, `docs/43` Sections 3.7 and 3.8)	`CV-` prefix	`CV-V1` to `CV-V7`, `CV-F1` to `CV-F7`	removes the collision of catalog F1 with stress-test F1 and configuration F
Paper-local catalog IDs	paper prefix, as NOTATION's "05-S2" rule	`02-G1` to `02-G9`, `06-C1` to `06-C4`, `08-T1`, `10-M0`	removes collisions with G1, G2, C1, C2 and configuration C
Published analyses (`docs/43` Section 3.1)	`PUB-` prefix	`PUB-01` to `PUB-05`	none in substance
Summary measures and time adjustments (`docs/43` S1 to S3, D1 to D7)	`SM-`, `TA-` prefix	`SM-1`, `TA-3`	removes collisions with sensitivity sweeps S1, S2 and with D_{it}
Exposure codings	E1 to E9, E9b, Encr2	unchanged	NOTATION extended E series
Constructs	"configuration A" to "configuration F", always spelled out, with ρ	"configuration C at $\rho = 0.7$"	never a bare letter

The registry of all IDs, with formula, track, source and stored-data key, is a machine-readable table (`docs/series/registry/analyses.csv`) from which Appendix A is generated.

2.8 Canonical names

- **The published process**: Hendrickson's `generateData()` and `buildtrialdesign()` at commit `58b32a9` (NOTATION Part 3). Never "the original code", "orig" or `42ac030` for the published process; `42ac030` is the initial commit, whose `generateData.R` equals `58b32a9`'s, but analysis defaults may differ (survey a, memory note).
- **The 2024 revision**: commits `8609f12`, `f6ee86d`, `f70f86d`, `325314f` (scale factor s_f, ratio coupling, timepoint-1 guard, `Dbc` in the analysis). **The vendored copy**: the 2024 state copied as `06dac83`, which `docs/54` calls `orig`.
- **The refined process**: as NOTATION Part 3; with the user decision of Section 2.11 its validated fixed-half-life analysis uses $t^a_{1/2} = 0.2$ week.
- **Configurations A to F**: as NOTATION Part 3, always with ρ. `covar` is replaced in prose by "configuration F at the package's parameters"; `orig` by "the published construct" or "the vendored copy", whichever is meant.
- **The published construct**: configuration A only (compound symmetry with $\rho_{cs} = 0.8$, constant cross block). AR(1) variants of the `04-power-simulation.R` construct are named by configuration (resolves CR-30).
- **Architectures**: mean-moderation, covariance-moderation, dual-channel. **Designs**: OL, CO, OL+BDC, Hybrid (capitalized).
- **The estimand**: the biomarker-treatment interaction $\beta_{bm:D}$ (docs/30 D3); "biomarker-by-drug" in the titles of 43 to 57 becomes "biomarker-treatment" (DQ-7, decided).
- **Participant**, never patient; **visit**, never occasion; **off-drug period**, never washout for the span after discontinuation.

2.9 Superseded material: the dated-note policy

The current documents carry dozens of inline "(Superseded 2026-10-NN …)" paragraphs; one paragraph about F1 recurs in six documents (survey c, cluster 14), and `docs/57` holds three successive recommendations. The series adopts a different rule.

1. **Chapters state the current answer only.** When a chapter is written, superseded statements are rewritten, not annotated. A chapter contains no dated correction paragraph at release.
2. **History lives in two appendices.** Appendix B (decision history) records each decision once: date, what was recommended, what replaced it, the evidence and the old document. Appendix C (errata) records each correction to an old document or a manuscript once, with its status.
3. **After release, changes are logged, not annotated.** Each chapter ends with a change log (date, section, change, reason). A dated inline note is allowed only where a result a reader may already have used changes sign or magnitude materially, and it points to the change log.
4. **Old documents are frozen.** Archived documents are not edited further, except for a banner at the top naming the chapter that replaces them.

2.10 Chapter template

Every chapter follows one template:

1. Title; timestamp line; **Author.** pmsimstats team; **Purpose** (question answered, audience); **Prerequisites** (chapters).
2. Contents block, as in this document.
3. Notation: local symbols only, with status; a pointer to NOTATION.
4. **Summary**: the results, each with its evidence label.
5. **Construct box**: a fixed table stating, for every numeric result in the chapter, the process (published, refined, configuration X at ρ), carryover (anchored or compounding), coupling rule, allocation (fixed or randomized), design, N (total), analysis ID, $t^a_{1/2}$ and n_{reps}.
6. Setting and argument (the body).
7. **Derivations**: one row per result with its statement, CM label and verification script.
8. **Numerical evidence**: tables with MCSE; each caption gives the script and output file.
9. **Relation to the manuscripts**: which manuscript sections depend on the chapter, and whether their numbers agree.
10. **Limitations and open items**, each linked to the open-items register of Appendix D.
11. **Reproduce**: commands, one per output.
12. References (from the series bibliography); change log.

2.11 User decisions of 2026-10-11

Three decisions were made while this plan was written and are treated as settled:

1. **Fixed analysis half-life.** The validated fixed-half-life fallback uses $t^a_{1/2} = 0.2$ week in both tracks, interaction and main effect. The value 0.5 week is described as the historical base setting of the stress test and the covariance study. Evidence for F1 at 0.2 week comes from the job 14 extension (F1h02: pooled null rate 0.050 over 12 non-prognostic cells, worst cell 0.058, worst prognostic cell 0.064; quoted, 57 Section 8.2). The 41-cell stress test (stored run and job 15) and the covariance study evaluated F1 and $D_{bc,it}$ at 0.5 week only (survey; observed for job 15's file layout), so chapters 20 to 22 must say which half-life each number used. Code and driver defaults set at 0.5 week (survey d, `docs/56` Section 8) are aligned in Phase 2 (Section 9).
2. **The older `.tex` whitepapers** (00 to 20, with 02, 09a and 19) are consolidated into one document that holds only results not covered by the newer whitepapers, with the suspect references checked or dropped; the originals then go to the archive. This document is Appendix I, the legacy record (Section 3.8).

3. **The `--extra` and `--extra2` arms** of the stress test are being rerun on the server mercury (jobs 21 and 22) to pair with job 15.

3. Target structure

The series has front matter, six Parts with 28 chapters, ten appendices and the mathematics compendium as Volume II. Lengths are estimates in letter-size PDF pages at the current rendering (for scale, `docs/37` is 1,673 lines and renders to 43 pages; observed). The total is about 670 pages, against roughly 50,800 lines of `.md` and `.tex` in `docs/` today (observed, `wc`). Much of the reduction in duplicated text is offset by tables that are now split across documents being given in full once. In the tables below, "Needs" lists the prerequisite chapters and "pp." the estimated length in pages.

3.1 Front matter

Table 6: Front matter of the series

ID	Title	Needs	pp.
`fm0`	Reading guide and index	none	6
`fm1`	Executive summary	none	4
`fm2`	Notation, vocabulary and conventions	none	8

`fm0`. Reading guide and index.

- Scope: three reading paths; status of every chapter; old-to-new map of every `docs/NN`.
- Sources: `docs/00`, `docs/23` Section 0, CM "Scope and reading order", Section 4 of this plan.
- New to write: all; generated from the registry `docs/series/registry/chapters.csv`.

`fm1`. Executive summary.

- Scope: the program's results in four pages: the published process and its defects, the refined process, the closed form, the recommended analysis, the main effect.
- Sources: `docs/50` (structure), 36 Section 1, 37 summary, 56 Section 1, 57 Section 1, NOTATION Part 2 (paper 01 at matched N).
- New to write: all, after Phase 1.

`fm2`. Notation, vocabulary and conventions.

- Scope: pointer to NOTATION; named processes, configurations, analysis IDs; evidence labels; Monte Carlo conventions; construct box.
- Sources: NOTATION Parts 1 to 3, `docs/29` (project terms), `docs/30`, Sections 2.5 to 2.8 of this plan, CM "Shared performance measures".
- New to write: the project glossary and the conventions.

3.2 Part I. The problem and the published process

Table 7: Part I of the series

ID	Title	Needs	pp.
ch01	The question: a predictive biomarker in aggregated N-of-1 trials	fm2	12
ch02	Trial designs, randomization paths and exposure clocks	ch01	14
ch03	The published process at commit 58b32a9	ch02	22
ch04	Three defects of the published process	ch03	18

ch01. The question: a predictive biomarker in aggregated N-of-1 trials.

- Scope: the estimand $\beta_{bm:D}$; predictive against prognostic; the sign convention; why carryover matters; a one-page history of the program.
- Sources: 24 "Big picture"; 40 Sections 1 to 2; 31 Part IV (Sections 14 to 16); 02 Sections 1 and 7; 21 "Biological Assumptions"; papers 01 and 06 introductions.
- New to write: the program history; a framing that answers the co-author's comments on 02 pages 2 and 3.

ch02. Trial designs, randomization paths and exposure clocks.

- Scope: OL, CO, OL+BDC, Hybrid, balanced placebo, cycle designs; schedules w_t; paths P; η_t; t_{od}, t_{sd}, t_{pb} and their restart rules; N as total and the 2026-08-08 matched-N correction.
- Sources: 13 Section 2 (after a check against 58b32a9); 33 Section 2.5 (design table); 40 Sections 2 to 3; 24 Component 2; 37 Section 2; 27 Sections 2a and 2c (cycle designs, Hybrid round trip corrected); NOTATION Part 2; CM "Trial designs and treatment exposure", "Time arguments of the components".
- New to write: one table per design; a Hybrid schedule figure.

ch03. The published process at commit 58b32a9.

- Scope: `generateData()` and `buildtrialdesign()` as published; parameters; compound symmetry $\rho_{cs} = 0.8$; constant cross block; step coupling; compounding recursion; silent repair; the published `lmer` analysis with binary D_{it}; the censoring grid; Figures 4 and 5; the commit attribution table.
- Sources: 40 Sections 3 to 6 and 8; 36 Section 3; 53 Sections 3 to 4; 55 Sections 3 to 4; 44 Section 2; 25 Section 4; 23 Section 3.1; CM "The published compound-symmetry construct" and the three "2024 revision" subsections; Appendix I Sections 1 to 3.
- New to write: the commit attribution table (42ac030, 58b32a9, 8609f12, f6ee86d, f70f86d, 325314f, 06dac83); what `implementations/original/` is and is not.

ch04. Three defects of the published process.

- Scope: matrices above the compound-symmetry ceiling, silently repaired; step coupling and the collapse under negligible carryover (attenuation, zero coupling gap); the compounding recursion (mean shortfall; negligible effect on power at 0.1 to 0.2 week).
- Sources: 36 Sections 1, 4, 5 (spine); 55 Sections 1, 5 to 8 (summary only); 53 Sections 1, 8 (summary only); 37 Section 5.1; 34 Section 4.6; 50 Section 1; 07 counts via Appendix I.
- New to write: one table reconciling the published power at $t_{1/2} = 0.1$ week (0.126, 0.078, 0.100, 0.076: cell, code, seeds), from 55 Section 7.3; one reference cell quoted throughout the series.

3.3 Part II. The data-generating process

Table 8: Part II of the series

ID	Title	Needs	pp.
ch05	Response components and the modified Gompertz trajectory	ch02	16
ch06	Carryover on the mean	ch05	10
ch07	Interaction architectures: mean, covariance and dual-channel	ch06	20
ch08	Biomarker coupling under carryover	ch07	12
ch09	The response covariance and the coupling ceiling	ch08	30
ch10	The refined process: specification, the adjustments evaluated, and the roadmap	ch04, ch09	18
ch11	Features of an N-of-1 data-generating process	ch10	26
ch12	Component decomposition, identification and reduction	ch05, ch07	22
ch13	Informative dropout and missingness	ch02, ch10	10
ch14	Beyond the refined process: a pharmacodynamic sensitivity model	ch09, ch10	10

ch05. Response components and the modified Gompertz trajectory.

- Scope: BR, PB, TV, BL_i; their clocks; $G(t_{od}; m, d, r)$, landmarks, conditioning; alternative families; paper 07's family result.
- Sources: 24 Components 1 to 3; 25 Sections 2 to 3, 5 to 8 (literature, alternatives, ranked impact); 40 Sections 3 to 4; CM "The response trajectory" (4 subsections); paper 07; 18 via Appendix I.
- New to write: a family-overlay figure; 25 Section 6 and Recommendation 4 rewritten to the corrected account of Section 3; the link from 25 Recommendation 5 to P04's $v_i \Psi_{it}$.

ch06. Carryover on the mean.

- Scope: decay families; the half-life as a pharmacodynamic response half-life; anchored against compounding carryover; restart on re-exposure; the 2024 scale factor.
- Sources: CM "Carryover on the mean" (3 subsections); 53 Section 5; 51 Features 3 to 4; 38 Sections 2 to 3; 40 Section 3; 24 Component 1.
- New to write: the reconciliation of a 2.3-hour plasma half-life with a 0.5 to 1 week carryover half-life.

ch07. Interaction architectures: mean, covariance and dual-channel.

- Scope: definitions; estimand and units per architecture; the latent-variable reading; biological rationale; paper 01's contrast; paper 11's dual channel.
- Sources: paper 01; CM "Two architectures", "The dual-channel architecture", "A latent-variable reading"; 21 "Biological Assumptions"; 23 Sections 1.6, 8.1 to 8.2, 11; 02 Section 4; 33 Sections 2.3 and 3.2; 36 configurations D and E; `plan-architecture-c.md`; Appendix I Section 5 (calibration).
- New to write: an estimand-and-units table; the current contrast (NOTATION Part 2; paper 01 under graded coupling) replacing 10 to 15% against 40 to 60%; the mean-moderation estimand shift (33 Section 2.3).

ch08. Biomarker coupling under carryover.

- Scope: coupling profile g_t; step, graded and ratio rules; the pharmacokinetic argument and its two qualifications; the coupling gap $\mathcal{G}$; shape of the decay.
- Sources: CM "The estimand and the coupling rules", "The pharmacokinetic argument for graded coupling", "ratio-graded", "Shape of the coupling decay"; 55 Sections 4 to 5; 37 Section 4.2; 51 Feature 5; Appendix I Section
  7.

- New to write: the answer to "is graded decay structurally required" (02 page 5, `notes.md`); the trait-against-state variance question as an open modeling choice.

`ch09`. The response covariance and the coupling ceiling.

- Scope: joint correlation matrix; the exact ceiling c^*_{bm}; the compound-symmetry construct, configuration F and the separable form; Kronecker decompositions; factorized ceiling and per-step costs; schedule dependence; silent repair; components and the ceiling; prognostic couplings in the ceiling; the factored generator; the outcome covariance.
- Sources: 35 "In brief" and Sections 4.1 to 4.8 (spine); 34 Sections 3 to 5; 36 Sections 4, 6.4, 6.5, 6.7, Appendix A; 37 Section 5.8; 32 Sections 3 to 5 (numbers labeled configuration F); 54 Section 5 (crossed decomposition, as corrected) if CM lacks it; 31 Section 20; 44 Steps 5 to 6; CM "Validity of covariance binding" (15 subsections).
- New to write: one full ceiling table (construct by $\rho \in \{0.7, 0.8\}$ by design by $t_{1/2}$); a box "a higher ceiling is not more power" (35 Section 6).

`ch10`. The refined process: specification, the adjustments evaluated, and the roadmap.

- Scope: the authoritative definition; the one table of differences from the published process; adjustments 1 to 3 and separability over configurations A to F; power under the adjustments; the step table of 44; a faithful published reference arm; operating characteristics of the refined process as a whole.
- Sources: 36 Sections 6.1 to 6.6, 6.8; 44 Sections 1, 3, 5, 6; 32 Section 2 (repair, choice, consequence); NOTATION "Named data-generating processes"; paper 01 `tab:spec`; 50 Section 1; 51 Section 20; 23 Section 2 (history only).
- New to write: the difference table; the refined-process operating characteristics from job 14 and job 15; the unreported common-grid contrast behind `36-fig4-contrast-reps100.png`.

`ch11`. Features of an N-of-1 data-generating process.

- Scope: the seventeen-feature framework and the five-DGP comparison.
- Sources: 51 Sections 1 to 19, 22 (spine); CM "Further features" (8 subsections).
- New to write: the evidence base stated per feature (the stress-test driver, now under the refined process).

`ch12`. Component decomposition, identification and reduction.

- Scope: identification of BR, PB, TV by the Hybrid design; three contrasts; the η limit; rank of the phase-augmented model; paper 06's studies; the two-component reductions.
- Sources: 24 identification and "Why the analysis does not match"; 31 Parts II, III (paper 06 studies, totals), IV, V; 28 Sections 3 to 4 (2026-10-08 corrections); CM "Partitioning the treatment effect" (4 subsections); papers 06, 13, 14.
- New to write: 31 Sections 18.1, 18.3 and 22.2 brought up to date (the prognostic couplings of the stress-test driver exist; F1 carries `De`).

`ch13`. Informative dropout and missingness.

- Scope: the dropout hazard; MCAR, MAR, MNAR; interaction with design; the stress test's MAR and MCAR mechanisms; the published censoring grid.
- Sources: CM "Informative dropout" (4 subsections); paper 09 (with the matched-N correction); 46 Section 5; 47 dropout cells; Appendix I Section 2.
- New to write: most of the chapter (no whitepaper covers dropout).

ch14. Beyond the refined process: a pharmacodynamic sensitivity model.

- Scope: turnover model; sensitivity binding; consequences for the test; a proposal, not implemented.
- Sources: 38 (spine, renamed $f(t) \to h(t)$, $F \to \Delta_h$); 44 Steps 7 to 8; CM "A pharmacodynamic sensitivity model" (4 subsections); 36 Section 6.7.
- New to write: σ_v^2 rewritten as τ^2 (DQ-6, decided).

3.4 Part III. Theory of the interaction test

Table 9: Part III of the series

ID	Title	Needs	pp.
ch15	Closed-form theory: E9, the coupling gap and power	ch08, ch09	30
ch16	Design efficiency and the choice of schedule	ch15	20

ch15. Closed-form theory: E9, the coupling gap and power.

- Scope: E9 and E9b; Propositions 1 to 3; power through the coupling gap; step coupling gives zero gap; graded coupling declines monotonically; anchored against compounding means; covariance binding as mean moderation; other designs; AR(1) structures; validation; the baseline row.
- Sources: 37 Sections 3 to 6 (spine); 41 Proposition 4 and Section 6; 39 Section 2; CM Part "Closed-form theory" (13 subsections); 17 via Appendix I (origin of E9).
- New to write: the reconciled anchored-change figures (CR-31).

ch16. Design efficiency and the choice of schedule.

- Scope: the criterion; contrast variance by design; switches, spacing, open-label visits, BDC and CO blocks; the exhaustive search; the design procedure; cycle count and period length.
- Sources: 39 (spine); 34 Section 5 (feasibility levers); 27 Section 4 (information formula, corrected optimum (2, 6)); CM "Design efficiency and time adjustment" (16 subsections); papers 05 and 12.
- New to write: the criterion renamed (39's $r^2_{\Delta B}$ is not a squared correlation); the construct stated in the opening.

3.5 Part IV. Analysis of the interaction

Table 10: Part IV of the series

ID	Title	Needs	pp.
ch17	The analysis model: exposure codings, time and baseline	ch15	22
ch18	Inference and calibration	ch17	16
ch19	The prognostic biomarker and the shared biomarker effect	ch17, ch18	16
ch20	The covariance study: size of candidate analyses	ch18, ch19	16
ch21	The stress test: protocol, results and the refined-process rerun	ch19, ch20	26
ch22	The recommended analysis	ch21	22
ch23	Choosing the analysis half-life from the data	ch22	30

ch17. The analysis model: exposure codings, time and baseline.

- Scope: the common model; E1 to E9 and the extended E series; the open-label branch; the pre-exposure value of $D_{bc,it}$; time adjustment (linear, spline, visit factor, phase); baseline as a row or a covariate (Propositions 1 to 5); cLDA; the published analysis term by term; why the analysis does not mirror the DGP.
- Sources: CM "The analysis model and its carryover specifications" (6 subsections); 43 Section 2; 40 Sections 8 to 9; 41 Propositions 1 to 5 and Section 7; 42 (pros, cons, update); 45 Section 7; 44 Section 5; 37 Appendix A "Visits before first exposure"; 24 "Why the analysis model does not match"; `plan-coherent-analysis-comparison.md` Q1 to Q3.
- New to write: one consolidated time-adjustment section (40 Section 9 and 42 merged).

`ch18`. Inference and calibration.

- Scope: what a random intercept absorbs under AR(1); a decomposition of realized Type I error; κ by design and architecture; Kenward-Roger and CR2 repair different failures; Satterthwaite degrees of freedom; the cost of dichotomizing.
- Sources: CM "Inference" (6 subsections); 45 Section 4; 33 Section 2.4; 56 Section 3.5; 36 Section 6.8; Appendix I Section 4 (the `lmer` evidence); Appendix H (RM-ANOVA).
- New to write: a κ section (33 Section 2.4 has the only analysis); the 0.14 to 0.16 against 0.14 to 0.23 ranges labeled by construct (CR-28).

`ch19`. The prognostic biomarker and the shared biomarker effect.

- Scope: the shared slope; the leak $(1 - f)s_{\text{prog}}$; freeing the baseline slope; the expectancy terms; the pathways table; leak factors; the power cost.
- Sources: 48 Sections 1 to 3 (lead-in); 49 Sections 2 to 4; 56 Section 3.3 (rigorous core); 45 Section 6; 41 Section 3; 51 Feature 6; CM "Baseline, a prognostic biomarker, and the shared biomarker effect" (6 subsections), "Prognostic pathways".
- New to write: F1 stated as approximately unbiased (CR-21); leak factors under the refined process if job 15 supports them.

`ch20`. The covariance study: size of candidate analyses.

- Scope: 63 analyses, 16 cells, 378,000 fits; size table; Kenward-Roger against CR2; size-adjusted power; the shared β_B; $D_{bc,it}$ as insurance; the manuscript 02 coding defect.
- Sources: 45 Sections 2 to 8 (spine); 43 Sections 3.7 to 3.8; the anchored main pass (`summary-alt250-null500-anchored.c` fixed allocation).
- New to write: the anchored-pass results; a statement of what the cancelled follow-up and $D_{bc,it}$ passes would have supplied.

`ch21`. The stress test: protocol, results and the refined-process rerun.

- Scope: criteria; the 14 analyses; the 41 cells; results of the stored run; results under the refined process (job 15, 11 analyses); G1, G2 and F1A paired with job 15 (jobs 21, 22); the symmetry screen; failed fits; the protocol of record as an annex.
- Sources: 47 (spine); 46 (annex, with Amendment 2 item 4 cut to a pointer); 48 Section 4; 49 Section 5; 50 Section 2; CM "The stress-test construct and its fourteen analyses" (5 subsections), "Size, power screens and the selection criteria".
- New to write: the job 15 results section; the job 21 and 22 results; the comparison of the two runs as independent proportions (`RERUN-anchored.md` Section 7).

`ch22`. The recommended analysis.

- Scope: F1 term by term; the fixed half-life of 0.2 week; worked examples; the decision framework; alternatives; the recipe in R; limitations; a summary of F1A0B with a pointer to `ch23`.
- Sources: 56 Sections 1 to 5, 7 to 8 (spine); 49 Section 6; 57 Sections 1 and 7.2 (summary); 57 Section 8.2 (F1h02 evidence).
- New to write: the status, half-life and limitations sections rewritten; Section 5 of 56 rebased on job 15 and job 14.

`ch23`. Choosing the analysis half-life from the data.

- Scope: why AIC with the interaction inflates size; what the data say; eight methods and permutation; worked examples; the decision table; the extension results; the cost.
- Sources: 57 (spine; Sections 6 and 7.3 moved to an annex); 47 Section 8; 56 Section 6; CM "AIC selection of the analysis half-life", "The assumed half-life of D_{bc}"; Appendix I Section 9 (`characterize_carryover`).
- New to write: F1A0, the bootstrap, max-$|t|$ and the orthogonality result moved into CM; the full-bootstrap check (GP-15) when run.

3.6 Part V. The treatment main effect and application

Table 11: Part V of the series

ID	Title	Needs	pp.
`ch24`	Testing the drug main effect when the half-life is unknown	`ch15`, `ch23`	30
`ch25`	Analyzing a real trial	`ch22` to `ch24`	12

`ch24`. Testing the drug main effect when the half-life is unknown.

- Scope: estimands β_D^{full}, β_D^{exp}, β_D^{bl}; probability limits and null bias; row-group weights; anchored against compounding; closed-form power; the half-life not identified under the null; max-$|t|$; randomization tests; averaging; joint estimation.
- Sources: 58 (spine); P04 (pointer for operations); CM "The treatment main-effect track" (to be rewritten); paper 04.
- New to write: the `M-` labels; the fixed 0.2 week policy stated for both tracks with 57's reconciliation of size (57 Section 1, Section 7.2 item 7).

`ch25`. Analyzing a real trial.

- Scope: `analyze_trial_extended()`, `characterize_carryover()`; predicted effect and threshold on the fitted (centered) scale; which visits carry the carryover signal; reporting (CENT 2015, PATH); two-stage random slopes for real data.
- Sources: 14 via Appendix I Section 9; vignette 03; 56 Section 8; 57 Section 5; 58 Section 10; 23 Section 10; 02 page 13 comment.
- New to write: most of the chapter.

3.7 Part VI. Notes for the original authors

Table 12: Part VI of the series

ID	Title	Needs	pp.
ch26	Note: the carryover recursion in the published code	none (standalone)	20
ch27	Note: step coupling in the published code	none (standalone)	24
ch28	Note: patching 58b32a9	ch26, ch27	8

ch26. Note: the carryover recursion in the published code.

- Scope: 53 as a standalone note.
- Sources: 53 (whole).
- New to write: deduplicate Section 8.3 and Section 9.5 against ch28.

ch27. Note: step coupling in the published code.

- Scope: 55 as a standalone note, plus the upstream never-treated Dbc coding defect.
- Sources: 55 (whole); 54 Section 3.6.2.
- New to write: the 54 Section 3.6.2 item.

ch28. Note: patching 58b32a9.

- Scope: the eight-line graded patch (power at 0.1 week from 0.126 to 0.638), the anchored fix, and their joint effect, with one power table.
- Sources: 37 Appendix A; 53 Section 9; 55 Sections 9 to 10 (Table 8, runs A to C); 36 patch material; the patch README under analysis/scripts/.
- New to write: the single power table.

3.8 Appendices and Volume II

Table 13: Appendices of the series

ID	Title	Sources	New to write	pp.
A	Catalog of analyses and the ID registry	43 (regenerated from analyses.csv, extended to 56 to 58)	the registry; the global IDs of Section 2.7	20
B	Decision history	the dated notes of 35, 41 to 45, 49, 50, 56, 57; 52 Section 2; 30 execution record; 45 Section 9; 57 superseded plans	all	8
C	Errata to earlier documents and manuscripts	36 Section 8; 40 Section 11; 45 Sections 8 and 10; CM main-effect section	the status of each item	8
D	Implementation status and open-items register	35 Sections 7, 9; 44 Section 4; 52 Section 4; 23 Section 14; 34 Sections 4.7, 6.2; 36 Section 9	one maintained register with one inspection date	8
E	Lexicon	29 (with the τ^2 entries fixed)	a project-terms table pointing to fm2	16
F	Style guide	30 (extended to the whitepapers); Sections 2 and 7 of this plan	the whitepaper scope	6
G	Bibliography	23 Section 17 (promoted); docs/references/*.bib	verification of the suspect entries (Appendix I Section 11)	14

ID	Title	Sources	New to write	pp.
H	Classical test equivalents: repeated-measures ANOVA	26 (whole; reserved letters renamed)	none beyond renaming	14
I	The legacy record: results of docs/00 to docs/20 not carried elsewhere	see the outline below	the consolidation (decided)	40
J	Software and parity workflow (pointer only)	22 (parity), moved to `analysis/scripts/parity/README.md`	a one-page pointer	1
Vol. II	Mathematics compendium	CM	updates of GP-28	(existing)

Outline of Appendix I, the legacy record (decided). One document that holds only what docs/00 to docs/20 (with 02, 09a and 19) contain and no newer whitepaper covers. Each section opens with a provenance line (source document, section, date, construct) and a status label; none restates a superseded mechanism except to name it as superseded.

Table 14: Outline of Appendix I, the legacy record (decided 2026-10-11)

Section	Content	Sources	Notes
0	Purpose, scope, and a map from each original to where its superseded content now lives	survey a recommendations	lists what was deliberately dropped (07a entire; 00, 01 entire; 03 and 04 mechanisms)
1	Commit history of the published code and the 2024 revision: per-commit agreement with published Figure 4 (r), PD failure rates (40 of 162 at publication, 102 of 162 after `8609f12`), condition numbers (median 229 against 3.1×10^{18}), eigenvalue range	04 Appendix A.7; 07 Section 3; 23 Section 3.1	commit attribution corrected to `58b32a9` against `42ac030` and the 2024 commits
2	The published simulation grid: the four `buildtrialdesign` calls, the censoring model and its five patterns, 72 cells by 1,000 replicates by 5 censoring conditions	13 Sections 2, 4, 6	each entry checked against `58b32a9`; "last observation carried forward" corrected to monotone dropout; dropout form checked against `censordata` (17 Section 7.1 item 4 disagrees)
3	Reproductions of published Figures 4 and 5	04 Section 2 ($r = 0.994$ at 200 replicates); 18 Sections 1 to 5	18's figures are missing; its comparison is reported as a historical result under the April construct, or regenerated (GP-26)
4	The analysis-model switch: `lmer` Type I error under AR(1) data (200 replicates; OL 0.13/0.09, CO 0.17/0.13 at two sizes) and `corCAR1` for unequal spacing	06 Sections 1, 3.3, 6	06's residual-correlation formula replaced by CM `eq:inva-ri`; "always understates" qualified

Section	Content	Sources	Notes
5	Mean-moderation formulations that failed (three attempts) and the calibration check (3.36 and 3.48 per SD against 3.6); positive definiteness under covariance moderation against mean moderation; the published grid under mean moderation	19.md (reviewed); 19.tex Sections 6 to 7	19.tex ceilings ("about 0.45") replaced by `ch09` values; the analysis variable of each table stated
6	The constant-exposure study and the two-period crossover bias	09a.md Sections 2 to 7 (corrected 2026-10-08)	the Shi and Ye (2024) expression with corrected signs; the withdrawn conclusion 4 stated as withdrawn
7	Open modeling questions: trait against state variance off drug; clinical plausibility of a negative BR under the step rule	08 Section 8; 09 Section 5	carried to `ch08` as open items
8	The co-author's ten questions on the April architecture paper, and where the series answers each	02 annotations (pages 2, 3, 5, 7, 13); `notes.md`	a table: question, answer, chapter
9	The March 2026 collaborator goals and their status: Goal 2b (residual analysis, never implemented); the threshold and predicted-effect formulas (defective, fixed in code 2026-10-08); `characterize_carryover()` grid (0.1 or 0.25 to 8 weeks, to be checked)	14; 03 Section 6	links to `ch23`, `ch25`
10	Abandoned designs and software history: the simplification plan (not adopted) as the origin of E9 and of the RM-ANOVA comparator; the 2025 tidyverse alignment and its unused decay functions	17 Sections 3.3, 8; 15; 16 item 1	one paragraph each
11	Bibliography corrections: the misattributions of 20 and their corrections; checks of Chen, Li and Yang (2020, *PLoS One* e0216949) and Eng and Konings (2018, *Trials* 19:72); also Pequignat et al. (2023) and De Bock et al. (2021)	20; 17 Section 7; 23 Section 17; 02 references	each entry verified against Crossref or PubMed or dropped (GP-29)

4. Disposition of every current file

"Archive" means `docs/legacy/` with the file name unchanged and a one-line banner naming the replacing chapter (Section 9.3). "Remove from series" means the file moves out of `docs/` (destination named) and is not part of the read-through. Where the surveys disagreed, the row says so and gives the resolution.

Table 15: Disposition of every file in `docs/` (survey recommendations; disagreements resolved in the rows for 09a, 19 and 32)

File(s)	Destination	Reason
00-documentation-index.tex	archive	superseded; replaced by the generated index `fm0`
01-codebase-overview.tex	archive	repository description belongs in `CLAUDE.md` and package docs; nothing unique

File(s)	Destination	Reason
02-dgp-mean-moderation-vs-mvn-rchcoments.pdf	`docs/reviews/`, renamed review-comments-2026-04-29.pdf; Appendix I Section 8	record of the co-author's comments; the questions enter Appendix I
03-audit-and-revision-report.tex	archive; Appendix I Sections 1, 9	claims superseded; corrected commit attribution in `ch03`
04-revised-power-analysis.tex	archive; Appendix I Sections 1, 3	commit history and Figure 4 reproduction are unique; mechanism and ceilings superseded
06-ar1-residual-correlation.tex	archive; Appendix I Section 4	`lmer` Type I evidence unique; derivation superseded by CM
07-positive-definiteness-failures.tex	archive; Appendix I Section 1	counts and condition numbers kept; root cause refuted by `docs/36` Section 8
07a-pd-constraints-theory.tex, .pdf	archive	superseded and partly wrong; nothing carried (surveys a and c agree)
08-biomarker-correlation-decay.tex	archive; `ch08` credit; Appendix I Section 7	origin of graded coupling; trait-against-state question unique
09-carryover-correlation-artifact.tex	archive; Appendix I Section 7	superseded by 55 and 36; clinical-plausibility point kept
09a-carryover-bias-variance.md, .pdf, .tex	archive all three; Appendix I Section 6	survey a offered "keep the `.md` as an appendix"; the user decision places its unique content in Appendix I; the `.tex` contradicts the corrected `.md`
13-figure4-walkthrough.tex	archive; Appendix I Section 2	grid and censoring specification unique; DGP sections mislabel the 2024 code
14-trial-analysis-implementation-guide.tex	archive; Appendix I Section 9; `ch25`	goals record; formulas superseded by the 2026-10-08 code fix
15-tidyverse-alignment-plan.tex	archive; Appendix I Section 10	planning record
16-tidyverse-code-review.tex	archive; Appendix I Section 10	planning record
17-simplification-plan.tex	archive; Appendix I Sections 10, 11	origin credit for E9 and RM-ANOVA; a suspect reference
18-response-parameter-sensitivity.tex	archive; Appendix I Section 3	figures missing; result kept as historical
19-mean-moderation-implementation-notes.md, .pdf, .tex	archive all three; Appendix I Section 5; `ch07` pointer	survey a recommended a merged appendix to the architectures chapter; the user decision places it in Appendix I
20-annotated-bibliography.tex	archive; Appendix G; Appendix I Section 11	misattributions; corrected list in 23 Section 17
21-architecture-comparison-summary.md, .pdf	archive; `ch07` (rationale)	superseded; 50-replicate tables unreproducible
22-journal-target-recommendation.md, .pdf	remove from series: the paper 01 folder, `planning/`	planning, not methods

File(s)	Destination	Reason
22-tidyverse-development-parity-workflow.md, .pdf	remove from series: a README in `analysis/scripts/parity/`; Appendix J pointer	current developer guide, not methods; frees the duplicate number
23-documentation-compendium.md, .pdf	archive; salvage 3.1 to `ch03`, 8.1 to 8.2 to `ch07`, 14 to Appendix D, 17 to Appendix G	every section has a fuller current home
24-component-decomposition-pedagogy.md, .pdf	`ch05` (components) and `ch12` (identification)	current pedagogy; analysis advice rewritten to point to `ch22`
25-gompertz-model-evaluation.md, .pdf	`ch05`	literature, alternatives and ranked impact kept; numerics from CM
26-rm-anova-interaction-test.md, .pdf	Appendix H	current derivation, reference character
27-cycle-period-design-sweep-plan.md, .pdf	remove from series: the paper 05 folder, `planning/`; Section 4 to `ch16`	planning record; corrected information result kept
28-hendrickson-hybrid-design-and-decomposition.md, .pdf	archive after porting its 2026-10-08 corrections to `ch12`	superseded by 31 (its own banner)
29-nof1-precision-medicine-lexicon.md, .pdf	Appendix E	reference; fix τ^2
29-nof1-precision-medicine-lexicon_2026-06-18_1719.pdf	delete	stale dated duplicate
30-terminology-consistency-plan.md, .pdf	Appendix F (extended); execution record to Appendix B	prose authority beside NOTATION
30-terminology-consistency-plan_2026-06-18_0834.pdf	delete	stale dated duplicate
31-component-decomposition-and-reductions.md, .pdf	split: Parts II to V to `ch12`; Parts I and VI removed (pointer to `ch05`); then archive	executes `plan-doc31-revision` Phase 0
32-why-covar-departs-from-orig.md, .pdf	`ch10` (framing), `ch09` (numbers labeled configuration F); then archive	survey b recommended merging into 54; 54 is itself largely superseded (survey d), so the framing goes to `ch10` instead
33-paper01-dgp-architecture-review.md, .pdf	remove from series: the paper 01 folder, `revisions/`; findings 2.3, 2.4 to `ch07`, `ch18`; Section 2.5 to `ch02`	manuscript review, not a methods chapter
34-cbm-feasible-range.md, .pdf	`ch09` (formula, schedule dependence), `ch16` (Section 5 levers); Sections 6.1, 7 archive; 6.2 to Appendix D	partly superseded at $\rho_{cs} = 0.7$
35-separable-response-covariance.md, .pdf	`ch09` spine; Sections 7, 9 to Appendix D	current
36-hendrickson-pd-and-carryover.md, .pdf	split: Sections 1 to 5 to `ch04`; 6.1 to 6.6, 6.8 to `ch10`; 6.7 and Appendix A.7 to `ch09`; 8 to Appendix C; 9 to Appendix D	current; drop "Working draft"
37-carryover-power-closed-form.md, .pdf	`ch15` spine; Appendix A to `ch28`	current
38-pharmacodynamic-carryover-model.md, .pdf	`ch14` spine	proposal; symbols renamed to 44's
39-nof1-design-optimization.md, .pdf	`ch16` spine	current within its construct

File(s)	Destination	Reason
40-hendrickson-hybrid-design-primer.md, .pdf	split: Sections 2 to 3 to `ch02`; 3 to 6 and 8 to `ch03`, `ch05`; 9 to `ch17`; 11 to Appendix C	primer; duplicated by 42
41-baseline-as-covariate.md, .pdf	`ch17` (Propositions 1 to 5); Proposition 4 to `ch15`; recommendation to Appendix B	partly superseded
42-flexible-time-adjustment.md, .pdf	`ch17` (time adjustment, merged with 40 Section 9)	duplicate tables removed
43-analysis-approaches-catalog.md, .pdf	Appendix A (regenerated)	reference, incomplete since 56
44-dgp-roadmap.md, .pdf	`ch10` (step table); Steps 7 to 8 to `ch14`; status to Appendix D	planning, current as status board
45-correctly-sized-hybrid-analysis.md, .pdf	`ch20` spine; Section 6 to `ch19`; 7 to `ch17`; 8 to Appendix C; 9 to Appendix B	covariance-study record
46-strawman-stress-test-protocol.md, .pdf	`ch21` annex (protocol of record)	historical protocol; Amendment 2 item 4 cut to a pointer
47-stress-test-results.md, .pdf	`ch21` spine	current record of the stored run
48-shared-biomarker-effect-and-stress-test-issues.md, .pdf	Sections 1 to 3 to `ch19`; 4 to `ch21`; 5 dropped	partly superseded; Section 4 table corrected (CR-20)
49-predictive-prognostic-biomarker-expectancy-term.md, .pdf	Sections 2 to 4 to `ch19`; 6 to `ch22`; 5 and 7 to Appendix B	recommendation superseded
50-dgp-and-analysis-status-brief.md, .pdf	archive; regenerated as `fm1`	superseded brief
51-ten-features-of-an-nof1-dgp.md, .pdf	`ch11` (renamed "seventeen features"); Section 20 to `ch10`; 21 to paper 01	current framework
52-whitepaper-review-2026-10-07.md, .pdf	archive; Section 2 to Appendix B; Section 4 to Appendix D	review log
53-carryover-recursion-in-published-code.md, .pdf	`ch26` (kept standalone, DQ-3)	current note for the original authors
54-covar-vs-orig-dgp.md, .pdf	archive; Section 3.6.2 to `ch27`; Section 5 to `ch09` if CM lacks it	largely superseded; assessment of the ratio rule overtaken by 55
55-step-coupling-in-published-code.md, .pdf	`ch27` (kept standalone, DQ-3)	current note for the original authors
56-recommended-analysis-expectancy-cr2.md, .pdf	`ch22` spine; Section 6 to `ch23`	current, internally stale in Section 9
57-data-chosen-carryover-half-life.md, .pdf	`ch23` spine; Sections 6 and 7.3 to an annex	current
58-main-effect-testing-and-carryover-half-life.md, .pdf	`ch24` spine	current
59-whitepaper-refactoring-plan.md, .pdf (this plan)	archive at release; its Section 4 becomes `fm0`'s old-to-new map	planning
plan-architecture-c.md	remove from series: the paper 11 folder, `planning/`	superseded as to placement
plan-coherent-analysis-comparison.md	remove from series: the paper 02 folder, `revisions/`; Q1 to Q3 to `ch17`	planning; the forbidden word on line 20 is left in the moved copy only
plan-doc31-revision.md, .pdf	archive	executed by this refactor

File(s)	Destination	Reason
notes.md	`docs/reviews/` with a header naming its source and date; Appendix I Section 8	reviewer comments
reorg-plan-2026-04-22.md	archive	executed in a different form
ZZCOLLAB_USER_GUIDE.md	remove from series: `docs/tooling/`	framework guide, no project content
pub_review_whitepaper_2026-08-16.md, pub_review_whitepaper_2026-08-20.md, pub_review_remediation_2026-08-20.md	`docs/reviews/`	manuscript reviews of the August campaign
Architecting_Precision_Medicine.pdf	moved to `docs/archive/`, labeled unreviewed (DQ-4, decided)	12.2 MB AI-generated deck, no text layer, cited nowhere, unreviewed
pmsimstats-preamble.tex	archive, beside the archived `.tex` set	needed to compile the archived `.tex`
figures/ (44 files)	stays; new figures named `chNN-figK-*.png`; `40-fig2` and `40-fig3` renamed to print order	archived documents keep their paths
archive/ (4 files)	stays, for non-whitepaper historical material; receives the slide deck (DQ-4)	already historical; the old whitepapers go to `docs/legacy/` instead
references/ (13 files)	stays; `nof1-simulation.bib` becomes the series bibliography source	`pmsimstats-nof1-whitepaper.md` there was not surveyed (Section 11)
reviews/ (1 file)	stays; receives 02's PDF, `notes.md`, the `pub_review` files	review records
.claude/	unchanged	settings, not documentation

5. Conflict register

Each conflict is listed once. "Best answer" names the source of the current answer; "Action" names where the series states it.

(Updated 2026-10-11, Phase 1: CR-31, CR-43, CR-44 and CR-51 are resolved in Section 5.2 by inspection and by comparing stored outputs, and GP-29 in Section 8. The evidence, with commands and outputs, is in the Phase 1 log `phase1-cr-log.md` in the working scratchpad (`SP/` below); it is to be carried into Appendix I. Corrections for archived documents (13, 17, 20, 23 and 03 in `docs/legacy/`) are recorded there, not made in the archived files.)

5.1 Resolved by later work: needs propagation only

Table 16: Conflicts resolved by later work (propagation only)

ID	Conflict	Documents and sections	Best answer (source)	Action
CR-01	Mechanism of the published collapse: standard-error inflation with an unbiased estimate, or attenuation	03 Section 4.2; 04 Section 3.3; 09 Sections 4.4, 6.3; 02 Section 6 against 36 Section 5.3, 37 Section 5.1, 55 Section 5	zero coupling gap; the estimate attenuates by about three quarters (36 Section 5.3; 55 Section 5)	`ch04`; old documents archived
CR-02	What the step rule does off drug	03 Section 2.2 ("correlation drops to zero instantly", lines 57 to 58, spot-checked) against 04 Appendix, 08 Section 1, 09 Section 3.1, 55 Section 4	full c_{bm} wherever the off-drug mean is nonzero (55 Section 4, code quoted)	`ch03`, `ch04`
CR-03	Cause and pattern of the published PD failures	03 Section 3.2; 04 Section 4.2; 07 Sections 3.4, 4.1 against 36 Sections 1, 4, 8	c_{bm} above the compound-symmetry ceiling 0.256 to 0.262; diagonal scaling cannot change definiteness (36 Section 8)	`ch04`; Appendix I Section 1 keeps the counts only
CR-04	Coupling ceilings	01 Section 6; 03 Sections 2.1, 2.7; 04 Table A.5; 07a Section 8; 19.tex Section 7.1; `CLAUDE.md` against 34, 35, 36, 52 Section 2	by construct: 0.256 (compound symmetry, $\rho_{cs} = 0.8$); 0.440 (separable, $\rho = 0.8$); 0.474, Hybrid 0.480 (separable, $\rho = 0.7$) (52 Section 2)	`ch09` full table; `CLAUDE.md` (Section 9.5)
CR-05	Size of the architecture contrast	02 Sections 3.1, 5.1; 19.md Background; 21; 23 Sections 1.4 to 1.6, 18; `CLAUDE.md` against NOTATION Part 2 and paper 01	Hybrid 6.9% (mean) against 15.4% (covariance) at matched total $N = 70$; under graded coupling losses nearly equal (NOTATION Part 2; paper 01 as restated)	`ch07`; `CLAUDE.md`
CR-06	Commit attribution of the scale factor, the timepoint-1 guard and `Dbc`	03 Sections 2.4, 2.5; 06 Section 1.1; 13 throughout against 04 Appendix, 36 Section 8, the memory note	2024 commits `8609f12` and `325314f`; the published analysis used `lmer` with binary D_{it}	`ch03` attribution table
CR-07	Whether ignoring serial correlation always understates standard errors; OLS inflating Type I error	06 Section 3.1; 17 Method 5 against CM `sec:inva-ri`	direction depends on the contrast; OLS is conservative for the within-participant interaction (CM)	`ch18`
CR-08	Residual correlation under random intercept plus AR(1)	06 Section 3.2 against CM `eq:inva-ri`	CM (rescaled by σ_ε^2)	`ch18`; Appendix I Section 4
CR-09	Threshold and predicted-effect formulas	14 Section 4.2 against the code fix of 2026-10-08 (52 Section 4)	computed on the fitted scale (code after 2026-10-08)	`ch25`; Appendix C

ID	Conflict	Documents and sections	Best answer (source)	Action
CR-10	Recommended interaction analysis	24 “Four reasons”; 26 Section 10; 28 and 31 Sections 12 to 13; 40 Section 10; 41 Section 8; 45 Section 9; 49 Section 7; 50 Section 2; 56; `CLAUDE.md` against 57 Sections 1, 7.2	F1A0B primary data-driven candidate, F1AB alternative, F1 at a fixed half-life the fallback (57 Section 1, lines 209 to 223, spot-checked); the fallback at 0.2 week (decided)	`ch22`, `ch23`; Appendix B
CR-11	Hendrickson et al. (2020) subtitle and author order; Haller and Ulm against “Eng and Konings” for one DOI; the other misattributions of 20	02 references; 20 Section 1; 17 Section 7 against 09a.md, 23 Section 17	“Statistical methods and theoretical findings”, Hendrickson first (PubMed-checked in 09a.md and 23 Section 17)	Appendix G; Appendix I Section 11
CR-12	Hybrid crossover blinded or open-label	31 Sections 3, 5, 7.1 against 24, 28 Section 3.1, NOTATION	blinded, $\eta_t = 0.5$ (NOTATION)	`ch02`, `ch12`
CR-13	Rank of the phase-augmented model	31 Sections 7.4, 8 against 24, 28 Sections 3.4, 4.2, CM “Rank of the phase-augmented model”	exact aliasing at every N; five identified parameters (24, 28, CM)	`ch12`
CR-14	$\mathrm{Cor}(D, \eta)$ on the Hybrid design	31 Section 11.3 (about +0.6) against 28 Section 4.3 note (0.509 Hybrid, 0.775 recovery design), 56 Section 3 (0.588)	the reproduced values of 28 and 56, with the design and row set stated	`ch12`, `ch19`
CR-15	Paper 06 Study A sample sizes	28 Section 4.3 (150, 100) against 31 Section 11.3 (300, 200)	totals (31; NOTATION Part 2 migration)	`ch12`
CR-16	Gompertz parameters per participant or population constants	28 Section 2.1; 31 Section 6.1 against 24, 25 Section 6, 31 Section 9	population constants with MVN deviations (24, 25)	`ch05`
CR-17	Whether a biomarker-TV coupling exists	31 Section 18.1 (“no biomarker-TV counterpart anywhere”, line 1837, spot-checked) against NOTATION ($c_{bm,TV}$), 46 to 49	exists in the stress-test driver (`c_tv`), not in the package	`ch12`, `ch19`
CR-18	Restoring the recursion in the package	32 Section 6 against paper 01	anchored carryover (paper 01; 32’s dated note)	`ch10`
CR-19	Mean moderation “immune to carryover”	21; 23 Sections 1.4 to 1.6, 18 against 33 Section 2.3 and paper 01	the estimand grows with carryover (13% Hybrid, 22% OL+BDC; 33 Section 2.3); losses not of the same kind	`ch07`

ID	Conflict	Documents and sections	Best answer (source)	Action
CR-20	ES against F1 on the non-prognostic cells	48 Section 4 table, row "Symmetry screen dropped" (line 384, spot-checked: "ES on non-prognostic cells") against 47 Section 7 and 46 Amendment 1	F1 ahead (0.242 against 0.278 maximin shortfall; 47 Section 7)	`ch21`
CR-21	F1 exactness under prognostic pathways	49 Section 1 item 3 against 49 Section 3 note, 47 Section 4, 56 Section 3.3	approximately unbiased; leak factors −0.051 to 0.171 (56 Section 3.3)	`ch19`
CR-22	Status statements: anchored reruns "pending", data-chosen half-life "not yet evaluated"	38 Section 7; 43 header; 44 Section 1, Step 3, Section 4; 45 header; 56 Section 9 (lines 1004, 1022, spot-checked); 57 Section 1 (line 144, spot-checked)	job 15 finished (stress test, refined process); covariance anchored main pass finished; follow-up and $D_{bc,it}$ passes cancelled; job 14 reported in 57 Section 8; jobs 21, 22 running (Section 1.1)	`ch20`, `ch21`, `ch23`
CR-23	Citation of the 14-analysis decision rerun	35 Section 10 ("Amendment 1") against 46 lines 401 to 447, 45, 52	Amendment 2	`ch09`
CR-24	Symbols of the pharmacodynamic model	38 ($f(t)$, F, ω_i, ζ) against 44 index (θ_i, κ, z; $h(t)$, Δ_h)	NOTATION: $h(t)$; 38's current ω_i, ζ, b_i	`ch14`
CR-25	Flexible-time remedy	40 Section 10 recommendation 2 (`varIdent` plus `corCAR1`, untested) against 42 update, 45 Section 4.4	CR2 removes the size excess (42 update)	`ch17`
CR-26	MCSE multiples of the flexible-time excess	40 Section 9 ("about 3 and 3.5") against 42 Section 6 (3.6 and 4.3, at the MCSE at 0.05)	42 (Section 2.6 convention)	`ch17`
CR-27	The 2024 ratio coupling rule	54 Sections 3.3.1 to 3.3.2, 6 ("Keep … the interaction gate replacement", lines 876 to 880, spot-checked) against 55 Sections 4.5, 9.4, 10	55: graded at half the stated rate, visit 1 never coupled, E9 power 0.495 against 0.721 at $t_{1/2} = 0$ (55 lines 444, 464, 685, spot-checked)	`ch08`, `ch27`; 54 archived
CR-28	The published analysis on AR(1) data in CO	43 Summary item 4 (0.14 to 0.16) against 43 O1 row and 36 Section 6.8 (0.14 to 0.23)	both correct for their constructs ($\rho = 0.7$ covariance study; $\rho = 0.8$ in 36)	`ch18` with construct boxes
CR-29	Two maximin readings	43 (non-prognostic cells: G2 0.232, F1 0.242) against 47 Section 7 and 52 (all 26 alternative cells: G1 0.308, G2 0.312, F1 0.358)	both correct; label the cell set	`ch21`
CR-30	"Published construct" for AR(1) variants	48 Section 3 note; 49 Section 3 note against NOTATION, 56 Section 3.3	configuration A only (NOTATION)	`ch19`
CR-56	Published power at $t_{1/2} = 0.1$ week	37 Appendix A (0.126); 51; 53 (0.078); 55 run C (0.100); paper 01, 50 (0.076, 0.08)	reconciled in 55 Section 7.3	`ch04` one table

ID	Conflict	Documents and sections	Best answer (source)	Action
CR-32	Lists of differences from the published process	23 Section 2 (seven); 32 Section 2 (four); 54 Section 1 (five); NOTATION; paper 01 Appendix B	NOTATION's definitions plus 53, 55 and 36; resolved in substance, never tabulated once	`ch10` one table (GP-02)
CR-33	What "original Hendrickson" means	28 Section 2; 31 Section 6 (`implementations/original/`, `58b32a9`); 54 (`orig` = `06dac83`) against 32, NOTATION, the memory note	the published process is `58b32a9`; `implementations/original/` is a corrected data.table reference; 54's `orig` is the vendored 2024 copy	`ch03`
CR-34	E2 dominance	27 Section 10 against paper 02 grid	paper 02 (27's note)	`ch16`
CR-35	Meaning of τ^2 in the lexicon	29 (between-patient variance; also "tau" for effect size) against NOTATION decision 1	NOTATION	Appendix E
CR-36	The main-effect model in CM	CM "The treatment main-effect track" (linear DGP, days, $\rho = 0.3$) against P04 and 58	P04 and 58 (refined process, weeks, $\rho = 0.7$)	CM update (GP-28); `ch24`
CR-37	Fixed analysis half-life policy	46, 47, 49, 50, 56, 57 (0.5 week); 56 Section 8 recipe (0.1 to 0.2 week); 57 Section 7.2 item 3 (0.2 no gain); 58 Section 12 and P04 (0.2 week, line 222 of 58, spot-checked); `CLAUDE.md` (0.5 to 1.0)	0.2 week in both tracks (decided); 0.5 week named the historical base setting	`fm2`, `ch22`, `ch24`; `CLAUDE.md`
CR-38	Hybrid round trip in the cycle-design plan	27 Section 2c against 24 Component 2, 33 Section 2.5	33 Section 2.5 table	`ch02`
CR-39	TV identification in the Hybrid design	24 Q&A and summary table ("off-drug timepoints with high efficiency") against 24 identification section	weakly identified; TV never observed without PB (24 identification section)	`ch12`
CR-40	The Gompertz rescale-factor account	25 Section 6 and Recommendation 4 against 25 Section 3 (corrected 2026-10-08)	numerator cancellation, `expm1` remedy (25 Section 3)	`ch05`
CR-41	Whether `lmer` was correctly sized for the published results	06 Section 7 ("correct under CS") against 18 Section 5.2 ("anti-conservative")	correctly specified for compound-symmetry data (06; inferred consistent with 36 Section 6.8, which finds excess only on AR(1) data); 36 open item 4 (OL+BDC under CS) remains open (CR-46)	`ch18`

5.2 Genuinely open: needs analysis, inspection or a decision

Table 17: Conflicts that need analysis, inspection or a decision

ID	Conflict	Documents and sections	Current best answer	Action
CR-31	Anchored-change figures inside 37	header note (closed form at most 0.017 in CO; Monte Carlo 0.021; lines 16 to 17, spot-checked) against Appendix A final bullet (closed form up to 0.041 in CO; Monte Carlo 0.018; line 1662, spot-checked); 39 header repeats 0.017	Resolved 2026-10-11: both closed-form figures are correct for different grids. Closed-form change in CO is at most 0.017 at the tabulated half-lives of 0 to 2 weeks (compound symmetry, mean moderation, E9, $t_{1/2} = 2$: 0.548 to 0.531) and 0.041 on the plotted curves to 4 weeks (same cell, $t_{1/2} = 4$: 0.505 to 0.464); Hybrid at most 0.0016, OL+BDC 0.0020. Monte Carlo change on the same seeds (carryover only): CO 0.018, Hybrid 0.003, OL+BDC 0.003. The header's 0.021 was a stored Monte Carlo change of a reseeded Hybrid cell (0.0216), not a carryover effect. 39's 0.017 is the CO change in `design-power.csv` (grid 0, 0.5, 1, 2), correct. Source: pre-anchored backups against the anchored outputs, and the recursive sandbox rerun on the same seeds (verified; `SP/phase1-cr-log.md`). 37 header and Appendix A and 39 header corrected with dated notes	quote in `ch15` with the grid stated (GP-34 done)
CR-42	Design power ranking	13 Section 10.1, 14 Section 6.3, 18 (Hybrid > CO > OL+BDC > OL) against 04 Section 5.3, 19.tex (Hybrid > OL+BDC > CO > OL at $c_{bm} = 0.45$) against the published abstract via 23 Section 17.1 and 17 Section 7.1 (CO above Hybrid)	depends on construct, c_{bm} and N allocation (inferred)	compute the ranking under the refined process at matched total N (GP-25); `ch02`, `ch16`

ID	Conflict	Documents and sections	Current best answer	Action
CR-43	Whether `characterize_carryover()` is the liberal F1A procedure, and its grid	03 Section 6.2, 14 Section 3, 23 Section 9.1 ("validated") against 57 (F1A liberal); 01 (0.1 to 8 weeks) against 03 Section 6.2 (0.25 to 8 weeks)	Resolved 2026-10-11: yes, it is an F1A-type procedure, and a weaker one. `characterize_carryover()` (`R/carryover_analysis.R` lines 27 to 132) fits `lme_analysis()` at each candidate half-life with the interaction in the model, picks the lowest AIC (lines 116 to 118) and returns the model-based t-test p-values of the drug effect and the interaction at that half-life as if it were fixed (lines 76, 90; printed at lines 318 to 325). Unlike F1A of 57 it compares REML likelihoods (`nlme::lme` default, `R/lme_analysis.R` lines 167 to 177), which are not comparable across half-lives because $D_{bc,it}$ changes the fixed-effect design; it uses model-based, not CR2, standard errors; and a failed `corCAR1` fit silently falls back to no residual correlation (lines 174 to 177), so the AIC can compare different covariance models. Default grid: 0.1, 0.25, 0.5, 1, 2, 4, 8 weeks (lines 28 to 29), so 01 is right and 03 Section 6.2 (0.25 to 8) is wrong; 0 is not in the grid. Its p-values are liberal (57, 58); "validated" in 23 Section 9.1 is not supported (inspected; code change proposed in `SP/phase1-cr-log.md`, not implemented)	`ch23`, `ch25`; Appendix D (code change)

ID	Conflict	Documents and sections	Current best answer	Action
CR-44	The published dropout form ($\beta_0 + \beta_1(\text{change})^2$)	13 Section 4 against 17 Section 7.1 item 4	Resolved 2026-10-11: neither is exact. At `58b32a9` (`R/censordata.R` lines 53 to 86, identical on GitHub and in the local copy) each participant-visit is flagged by two independent uniform draws: a flat flag with probability $\min\{1,\ f_1\,\Delta t_k/\overline{\Delta t}\}$ and a biased flag with probability $\min\{1,\ f_2\,p_{ik}/\bar{p}\}$, where $p_{ik} = \Delta t_k\,(D_{ik}+100)^{e_{b1}}$, Δt_k is the visit interval (`t_wk`), D_{ik} the summed component improvement $TV + PB + BR$ at visit k (`D_` columns, `generateData.R` line 174), $f_1 = \beta_0(1-\beta_1)$, $f_2 = \beta_0\beta_1$, and the bars are means over all participant-visits of the trial; a visit is dropped if either flag is set, and dropout is carried forward (monotone). 13's $(\text{change}+100)^{e_{b1}}$ term is right but its direction is wrong: because D_{ik} is improvement, participants who improve more drop out more, contrary to the roxygen text and to 13; 17's "responders more likely to drop out" has the direction right but the form wrong (no additive β_0, no square of the change alone). β_0 is the per-visit flag rate, not the total dropout (inspected; direction and per-visit rate verified on synthetic data, `SP/cr44-censor-check.R`)	inspect `censordata` at `58b32a9` (GP-36); Appendix I Section 2, `ch13`
CR-45	σ_v^2 against τ^2	38 and 51 (v_i with σ_v^2) against NOTATION, P04, 58 ($\text{Var}(v_i) = \tau^2$)	one concept (DQ-6, decided): τ^2 everywhere	NOTATION revision; `ch11`, `ch14`
CR-46	Unexplained OL+BDC Type I error under compound symmetry	36 Section 9 item 4; 37 Section 6	open	analysis (GP-32)
CR-47	Silent repair still in the package sampler	23 Section 18 ("zero silent PD corrections") against 34 Section 4.7, 52 Section 4 (`R/generateData.R:391-393`), 51 Section 20	the Cholesky fallback still repairs silently (inspected by 34 and 52; not re-checked)	code fix; Appendix D

ID	Conflict	Documents and sections	Current best answer	Action
CR-48	Parity report mode	22 (parity) Section 9 against the committed quick-mode `parity_report.rds` (52 Section 4)	mismatch stands (52; not re-checked)	regenerate in default mode; Appendix D
CR-49	Manuscript 02 pre-exposure coding of $D_{bc,it}$ and its CO rerun	45 Sections 8, 12 question 3; 44; 52	defect confirmed (45 Section 8: CO power 0.484 to 0.744 after correction, compound symmetry); rerun of paper 02 unconfirmed	rerun (GP-31); Appendix C
CR-50	Driver cross-component values in the April comparison; the 19 against 21 Hybrid row	21 (0.1, 0.05) against 23 Section 6.1 (0.2); 19 (0.74/0.68/0.72) against 21 (0.74/0.72/0.68)	not resolvable (driver archived, output missing)	drop the tables; mention in Appendix I Section 5 only
CR-51	Failed-fit count and DS shares	48 Purpose (27 failed fits in 291,500) against 47 Section 9 (sum 28); 50 Section 2 (DS: 82%, 86%) against 56 Section 6.1 and 57 Section 3.5 (F1A: 87%, 77%)	Resolved 2026-10-11: no statement is wrong; they use different denominators and analyses. Failed fits: 27 of 291,500 is the 39-cell run of 48 (26,500 trials by 11 analyses); the final 41-cell main run has 28 of 308,000 (the two `prog-BL` cells add one F1 failure); 47 Section 9's per-analysis counts are exact over 14 analyses and 28,000 trials each (55 of 392,000; at most 0.6% per cell, G1 and G2 in cell 1). Selection shares on the 1,000 null trials of each cell: DS (S with AIC) chose 0 in 81.6% at true $t_{1/2} = 0$ and 1 or 2 weeks in 85.7% at true 1 week (50's 82%, 86%); F1A chose 87.0% and 76.7% (56's and 57's 87%, 77%). Source: `replicates-full.rds`, `-extra.rds`, `-extra2.rds` (verified)	state analysis and denominator in `ch21`, `ch23`

5.3 Needs the pending reruns, or the reporting of finished ones

Table 18: Conflicts that wait on reruns or on reporting finished runs

ID	Conflict or open number	Documents and sections	State on 2026-10-11	Action
CR-52	Cost of F1 relative to S	47 Section 1, 48 Section 3, 49 Section 1.5, 56 Section 1 (29%, stored run) against 57 Section 8.3 (F1 0.438 under the refined process)	job 15 holds S and F1 under the refined process (observed, 451 rows); not yet tabulated	report in `ch21`, `ch22` (writing only)
CR-53	G1, G2 and F1A under the refined process	46 Amendment 2, 47 Section 7, 49 Section 5	local anchored arms unpaired with job 15; jobs 21 and 22 running (decided)	report when jobs 21 and 22 finish; `ch21`
CR-54	The covariance study under anchored carryover	45 (recursive, fixed allocation); 35 Section 10	anchored main pass finished (320 rows, observed); follow-up and $D_{bc,it}$ passes cancelled	report the main pass in `ch20`; state that the follow-up and $D_{bc,it}$ passes stay cancelled and that 45 Sections 6, 7 rest on the recursive passes (DQ-8, decided)
CR-55	The 41-cell selection under the refined process	46 Amendments 1, 2; 47 Section 7 (F1 by the tie rule)	needs job 15 plus jobs 21, 22 through `12-strawman-stress-decision.R --tag -anchored`	rerun the decision script when 21 and 22 finish; `ch21`

6. Redundancy map

Table 19: Redundancy clusters and their single home

Cluster	Documents now	Single home	What the others become
R-01 Description of the published process	24, 25 Section 4, 28 Section 2, 31 Sections 2, 6, 36 Section 3, 37 Section 2, 40 Sections 2 to 6, 41 Section 3, 42 Section 2, 44 Section 2, 53 Section 3, 13	`ch03`	pointers; glossaries of 53, 55 to 58 point to `fm2`
R-02 The April audit (what changed relative to the published code)	03 Section 2, 04 Section 3 and Appendix A, 13 scope box, 14 Section 4, 23 Section 2, CM 2024-revision subsections	`ch03` (attribution) and Appendix I Section 1 (per-commit numbers)	archived
R-03 Differences between the package and the published process	23 Section 2, 32 Section 2, 33 Section 3.4, 54 Section 1, 53, 55, 36, NOTATION	`ch10` difference table	archived or pointer

Cluster	Documents now	Single home	What the others become
R-04 Positive definiteness and the ceiling	01 Section 6, 03 Section 3, 04 Section 4 and Table A.5, 07, 07a, 19.tex Section 7, 23 Section 3, 31 Section 20, 32 Sections 3.3, 4.3, 8, 34, 35, 36 Sections 4, 6.5, Appendix A.7.4, 37 Sections 5.5, 5.8, 44 Step 6, 46 Section 3, 50 Section 1, 51 Feature 5c, 52 Section 2, 54 Sections 4 to 5, 55 Section 8.2, CM	`ch09`; derivations in CM	removed; one-line pointers
R-05 Kronecker decompositions and generative forms	35 Sections 4.2, 4.4 to 4.7, 36 Sections 6.4, 6.7, Appendix A.7.5 to A.7.6, 37 Section 5.8, 38 Section 5, 44 Step 5	`ch09`	removed
R-06 Step coupling and the collapse	04 Section 3, 08 Sections 1, 5, 09, 09a.md Section 5.5 note, 13 scope box, 23 Section 5.7, 34 Section 4.6, 36 Section 5, 37 Sections 5.1 to 5.4 and Appendix A, 40 Section 5, 44 Step 2, 50 Section 1, 51 Feature 5, 55, CM "Step coupling"	`ch04` (diagnosis), `ch27` (note)	others archived
R-07 Graded coupling from a pharmacokinetic argument	04 Section 3.5, 08, 14 Section 3.1, 23 Section 5, 55 Section 3, CM `sec:inva-pk`	`ch08`	08 credited, archived
R-08 The compounding recursion	08 Section 2.2, 09 Section 2.2, 13 Section 3.4, 51 Feature 3, 53, 54 Section 3.5, CM "Anchored versus compounding"	`ch06` (mean), `ch26` (note)	removed
R-09 The two power tables of 53 and 55	53 Section 8.3 and Table 8, 55 Table 8 and Section 10 Table 11	`ch28` (one table)	53 and 55 point to it
R-10 Mean against covariance moderation	02, 19 (.md and .tex), 01 Section 1, 03 Section 5.4, 21, 22 (journal), 23 Sections 1, 8, `plan-architecture-c.md`, 33, 36 configurations D and E, paper 01, CM Part II	`ch07`; Appendix I Section 5 (history)	archived
R-11 Gompertz curve and parameters	23 Section 2.6, 24, 25, 28 Section 2.1, 31 Section 2, 40 Section 4, CM	`ch05`	removed; three notations reduced to m, d, r
R-12 Component decomposition pedagogy	24 (whole) = 31 Parts I, III Section 8, VI; 40 Sections 3 to 4, 11; CM; paper 06	`ch05` and `ch12`	31 Parts I and VI deleted
R-13 Hybrid design, schedule and expectancy	24 Component 2, 27 Section 2c, 28 Section 3, 31 Sections 5, 7, 33 Section 2.5, 37 Section 2, 40 Sections 2 to 3, glossaries of 53, 55 to 58, CM	`ch02`	pointers
R-14 Paper 06 results	28 Section 4, 31 Section 11, paper 06	`ch12`	28 archived
R-15 Pharmaco-dynamic carryover and the response half-life	38, 44 Steps 7 to 8, 40 Section 4, 39, 38 Section 2 = 40 Section 3 = 45 Section 7 (prazosin)	`ch06` (half-life), `ch14` (model)	removed
R-16 E9 and E9b closed form	17 Section 3.3, 37 Sections 3 to 6, 39 Sections 2 to 3, 40 Section 8, 41 Proposition 4 and Section 6, 43 Section 3.5, CM	`ch15`	removed

Cluster	Documents now	Single home	What the others become
R-17 Design efficiency and cycles	23 Section 13.1 item 4, 27 Section 4, 34 Section 5, 39, CM	`ch16`	removed
R-18 Analysis model and its evolution	23 Section 4, 24, 25 Section 6b, 26 Section 9, 27, 28, 31, 06, 43, 45 to 49, 56, 57	`ch17` (model), Appendix B (evolution)	removed
R-19 Time adjustment	40 Sections 9 to 10, 42 (whole), 45 Section 4.4, 43 time-adjustment rows	`ch17`	42's tables and figures not duplicated
R-20 Baseline coding	41, 43 Section 2.5, 45 Sections 6, 10, CM	`ch17`	removed
R-21 Alternative analyses and test methods	02 Section 6, 17 Section 3.3, 23 Sections 10, 13, 26, 43, `plan-coherent-analysis-comparison.md`, CM	Appendix A (catalog), Appendix H (RM-ANOVA), `ch17`	removed
R-22 Shared slope and prognostic leak	41, 45, 48 Sections 1 to 3, 49 Sections 2 to 3, 51 Feature 6, 56 Section 3.3, CM	`ch19`	48 Sections 1 to 2 kept as the lead-in only
R-23 Expectancy-drug confounding	40 Section 8, 49 Section 2, 51 Feature 1, 56 Section 3.3, 58 Section 6, CM	`ch19`	removed
R-24 Stress-test design and results	46, 47, 48 Section 5, 49 Sections 1, 4, 5, 50 Section 2, 56 Sections 5, 7, CM	`ch21`	46 as annex; others removed
R-25 Symmetry screen	46 Amendment 1, 47 Section 5, 48 Section 4, CM "Sign symmetry of power"	`ch21` (one section)	removed
R-26 AIC choice of the half-life	47 Section 8, 49 Section 5, 50 Section 2, 56 Section 6, 57, 58 Section 9, CM	`ch23` (interaction), `ch24` (main effect)	removed
R-27 Recommendation and recipe	49 Section 7, 50 Section 2, 56 Sections 7 to 8, 57 Section 7.2	`ch22`	removed
R-28 "Superseded by F1" notes	35 Section 10, 41 Sections 1, 8, 42 update, 43 Sections 1, 5, 44 Sections 1, 5, 45 header and Sections 1, 9	Appendix B (one entry)	removed
R-29 Package implementation status	34 Section 4.7, 35 Sections 9, 10, 36 Sections 6.7, 9, 44 Sections 3, 4, 50, 51 Section 20, 52 Section 4	Appendix D	removed; chapters link to it
R-30 Repository, parity and tooling history	00, 01, 03 Section 5, 15, 16, 22 (parity), 23 Sections 7, 14, 15, `CLAUDE.md`	`CLAUDE.md` and `analysis/scripts/parity/README.md`; Appendix I Section 10	archived
R-31 Bibliography	20, 23 Section 17, reference lists of 02, 09a, 17 and of 34 to 58, `docs/references/`	Appendix G and one `.bib`	per-chapter lists generated from the `.bib`

Cluster	Documents now	Single home	What the others become
R-32 Terminology authorities	NOTATION, 29, 30, `CLAUDE.md` lexicon section, every notation index	NOTATION (symbols), `fm2` (project terms), Appendix E (lexicon), Appendix F (style)	`CLAUDE.md` section points to them
R-33 Main-effect half-life mathematics	58, P04 Revision 1 tables, CM main-effect section	`ch24`; P04 keeps operations only	P04 points to `ch24`

7. Notation and convention fixes

7.1 Documents before 34: wholesale

These documents are archived rather than edited (Section 2.9), so the fixes apply when their content is carried into a chapter or into Appendix I.

Table 20: Notation and convention fixes for content carried from documents before docs/34

Fix	Where it occurs (survey)	Canonical form
Architecture letters	all `.tex`; 21, 23, 32, `plan-architecture-c.md`	mean-moderation, covariance-moderation, dual-channel
"N-of-1" for the Hybrid design	13, 14, 18	Hybrid
ER, TR, "pharm_biomarker" for components	07, 07a, 09, 13, 15, 18	PB, TV
$S_{i,t}$, `Sx` as a symbol	`.tex` set	Y_{it}
ρ for compound symmetry	07a, 18, 23 Sections 2.1 to 3.3, 31 Section 20, 32 (ρ_c)	ρ_{cs}
ρ or ρ_t for the biomarker-response correlation	04 Section 3.1, 08, 09	$c_{bm}\, g_t$
c_{bm} as the mean-moderation multiplier	01, 19.tex	β_{bm}
B_i^* for the standardized biomarker	01, 19.tex	b_i
D_{bc} for a binary indicator	06 Section 4.1, 09	D_{it}
c_{tv}, c_{pb}, c_{br} for autocorrelations; c_{cf1t}, c_{cfct}	07, 07a, 24, 31	ρ_{cs} or ρ; c_1, $c_\times$
e_t for expectancy	07	η_t
α for the decay factor or a power-law exponent; λ_{cor}, λ_{drug}	08, 09	ϕ_t; λ

Fix	Where it occurs (survey)	Canonical form
f_G, a, g_b, g_c for the Gompertz curve	08, 25	$G(t_{od}; m, d, r)$
β_t, β_1, β_2 for time and biomarker coefficients	23 Section 4.1, 24, 25, 31	β_T, β_B
K for blocks, visits or components; n for occasions	17, 23 Section 13, 24, 31 Section 20	n_c; n_{on}, n_{off}; K only for the cross-component matrix
g, ϕ_j, τ_t in RM-ANOVA	26	another stratum letter; ω_j or similar for phase effects; another letter for time effects
v_i for a random slope	24	another letter (v_i is the drug-effect deviation)
$\tilde{r}$, v, r for the coupling vector	31, 32, 34, 36	$c_{bm}\tilde{g}$, $\tilde{g}$
g_t as the off-drug factor only	32	g_t is the whole profile (1 on drug)
$N(\cdot)$ for the normal law	09a.md, 25, 26	$\mathcal{N}$
local labels for now-canonical symbols (ϕ_t, g_t, μ_{bm}, w_t, c_1, M, m, d, r)	23, 24, 32, 33	marked canonical
"biomarker-by-treatment", "washout" for the off-drug span, lower-case "hybrid"	21, 23, 24, 28, 31	biomarker-treatment interaction; off-drug period; Hybrid
hand-written "Rendered on" footers with `~/prj/alz/...`	every `.tex`; `reorg-plan-2026-04-22.md`	removed; `tools/render.sh` owns provenance
British spellings	`reorg-plan-2026-04-22.md`	US English

7.2 Regressions in 34 to 58

Table 21: Notation regressions in docs/34 to docs/58

Document and location (survey)	Problem	Fix
34 Sections 3.1 to 3.2	$\tilde{r}$; unpadded g in the ceiling formula; C as cross block	$\tilde{g}$; $(\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$; name the block explicitly
35	q_K, q_A (eigenvectors); Δ as a time lag; μ as an eigenvalue; v_{TV}, v_{PB}, v_{BR}; s_t as factor SDs; $c_{bm,BL}$ marked local	letters chosen against NOTATION's reserved list; $c_{bm,BL}$ marked canonical
36 Section 6.7, Appendix A.7.5	q for the quadratic form; r for the coupling vector; K, A_c marked local	another letter; $c_{bm}\tilde{g}$; canonical
37 Sections 4, 4.4 (lines 492, 624, 635)	plain $E(\cdot)$; μ, μ_p, $\bar{\mu}$ local; v_t factor SDs; e_i residual; "patient(s)"	$\mathbb{E}$; another letter; participant
38	$f(t)$, $\bar{f}$, F; G, H in the Section 6 proof; σ_v^2	$h(t)$, Δ_h (as 44); other letters; τ^2 (DQ-6)

Document and location (survey)	Problem	Fix
39	$f(d)$; ω (measurement-error share); C (response covariance); v; d (spacing); w_t marked local; $r^2_{\Delta B}$ for a non-correlation; "Values changed by at most 0"; "patient(s)"	other letters; rename the criterion; fix the phrase; participant
41 line 205	$N(\cdot)$; $f(t)$; d_{it}	$\mathcal{N}$; another letter; another letter
42, 45	$f(t)$ for the time adjustment	another letter (for example $s(w_t)$)
44 index	stale description of 38's symbols	NOTATION's
45 index	B_i "centered"	B_i^c
48 Section 3 note; 49 Section 3 note	"published construct" for AR(1) variants	configuration name with ρ
49, 56	n_{on}, n_{off}, n_v (49) against n_0 to n_3 (56)	NOTATION's n_{on}, n_{off}, n_v
51	σ_v^2; $E(t)$, E_{it} beside E1 to E9; L, U beside L_{it}; file name "ten"	τ^2 (DQ-6); other letters; rename
53	m (run position), L (run length)	other letters
54	w_t marked local; "occasion"; D_{bc} without subscripts; n for occasions	canonical; visit
56 Section 1	AR(1) exponent on the visit index	$\lvert w_t - w_s \rvert$
57	M, m (bootstrap); $g(Y)$; γ; h (candidate half-life); ω ($\log t^a_{1/2}$); $\hat{\xi}$; e_i; grid $\{0, 0.25, 0.5, 1, 2\}$	letters chosen against NOTATION's reserved list (for example R resamples, $\mathcal{S}(Y)$ for the selection rule); one grid across 57 and 58 (DQ-10)
58	B (GLS bread), G (randomization distribution) beside the Gompertz G, $\hat{K}$, Z, r_p, f_m, M, h; grid $\{0, 0.2, 0.5, 1, 2\}$; analysis labels shared with the interaction track	other letters; `M-` labels (Section 2.7)

7.3 Cross-cutting rules for the series

1. NOTATION is the only symbol authority; a chapter adds only local symbols, never on a reserved letter.
2. Every result states its construct in the chapter's construct box (Section 2.10); a ceiling, size or power number without ρ, the construct, carryover, coupling, allocation and N is not quoted.
3. N is the total; $t_{1/2}$ in weeks; the fixed analysis half-life is 0.2 week unless a historical result is quoted (decided).
4. Evidence labels from Section 2.5 only.
5. Analysis IDs from the registry only (Section 2.7).
6. Each table has a caption line, each figure a caption with its generating script; figure files named in print order.
7. No dated correction paragraphs at release (Section 2.9).
8. US English; no em dashes; prose vocabulary per Appendix F; this project's studies are exploratory and its protocols are working protocols, never described with the vocabulary of registered trials.
9. The linter `tools/notation-lint.pl` runs over every chapter; two checks are added for the series: plain $E($ and $N($ for the normal law, and reserved letters reused in a chapter's local table (GP-37).

8. Gaps

(Updated 2026-10-11: GP-29, GP-34, GP-35 and GP-36 are done; see the rows and Section 5.2.)

Merged from the four surveys' gap lists and deduplicated. "Writing" means the content exists in outputs or derivations and needs only to be written; "computation" means new runs or code; "blocked" means it waits on a running job or a decision. Costs are rough working time with assistance; server time is stated separately.

Table 22: Gaps the series must fill

ID	Gap	Status	Rough cost
GP-01	Front matter: reading guide, executive summary, project glossary	writing	1 day
GP-02	One table of differences between the published and the refined process	writing	0.5 day
GP-03	Commit attribution table (`42ac030` to `06dac83`), checked by `gh api`	writing, light inspection	0.5 day
GP-04	One design table per design, including balanced placebo and cycle designs; Hybrid schedule figure	writing	1 day
GP-05	The full ceiling grid (construct by ρ by design by $t_{1/2}$)	computation, closed form (existing `cbm-ceiling` scripts)	2 hours
GP-06	Estimands and units per architecture; the mean-moderation estimand shift	writing (33 Sections 2.3, 3.2)	0.5 day
GP-07	κ by design and architecture	writing from the 33 Appendix 1 reanalysis	0.5 day
GP-08	Plasma against pharmacodynamic half-life	writing	2 hours
GP-09	Answers to the co-author's ten questions	writing	2 hours
GP-10	Trait against state variance off drug, as an open choice	writing	2 hours
GP-11	Operating characteristics of the refined process as a whole	writing from jobs 14 and 15	1 day
GP-12	Stress test under the refined process, all 14 analyses	blocked on jobs 21, 22 for G1, G2, F1A; job 15 part is writing	1 day after the jobs
GP-13	Covariance study under anchored carryover	writing (main pass); the follow-up and $D_{bc,it}$ passes stay cancelled (DQ-8, decided)	0.5 day
GP-14	F1 cost relative to S under the refined process	writing (job 15)	2 hours
GP-15	Full bootstrap ($M = 999$) check of F1A0B in a few cells	computation: about 17 worker-minutes per trial (57 Section 8.4), so 3 cells of 200 trials is about 170 worker-hours	server
GP-16	Sample-size planning for F1 at 0.2 week and F1A0B (power by N, ρ, c_{bm})	computation	2 days plus server
GP-17	The interaction analysis in CO and OL+BDC (expectancy terms constant or confounded differently)	computation	2 days plus server
GP-18	Prognostic couplings at strengths other than 0.3 and combined pathways	computation	1 day plus server
GP-19	Simulation of Features 11 to 17 of 51	computation (large)	defer
GP-20	Calibration of effect sizes and half-life to prazosin trial data	blocked on data access	unknown
GP-21	Main-effect stress test under the rebuilt P04	computation (paper 04's plan)	per P04 Section 7.3
GP-22	Coverage of profile-likelihood and PoSI-type intervals	computation	1 day plus server

ID	Gap	Status	Rough cost
GP-23	Monte Carlo methodology section	writing	0.5 day
GP-24	Dropout chapter under the refined process	writing (CM, paper 09, stress-test MAR and MCAR cells)	1 day
GP-25	Design ranking under the refined process at matched total N	computation (closed form plus a small Monte Carlo)	1 day
GP-26	Figures: Gompertz family, Hybrid schedule, coupling profiles, ceiling against ρ; optional Figure 5 regeneration	writing and plotting from existing scripts	1 day
GP-27	Implementation status appendix (separable `buildSigma`, factored generator, Cholesky fallback, `cbmCeiling()`, anchored carryover, prognostic couplings in the package)	writing; the code work is separate	0.5 day
GP-28	CM updates: main-effect track (version 1 to P04 and 58); 57 procedures; 55 cited; anchored stress-test DGP; 58 Propositions 1 to 7	writing (LaTeX)	2 days

ID	Gap	Status	Rough cost
GP-29	Verification of four suspect references (Chen, Li and Yang 2020; Eng and Konings 2018; Pequignat et al. 2023; De Bock et al. 2021)	done 2026-10-11 (PubMed and Crossref): all four misattributed, none exists as cited. "Chen, Li and Yang": the title is Blackston, Chapple, McGree, McDonald and Nikles (2019), *Healthcare* 7(4):137, doi:10.3390/healthcare7040137; the cited DOI 10.1371/journal.pone.0216949 is Ao et al. (2019), an Ebola-glycoprotein HIV vaccine paper, *PLoS One* 14(5). "Eng and Konings": the title and DOI are Haller and Ulm (2018), *Trials* 19:128 (not 19:72), doi:10.1186/s13063-018-2491-0. "Pequignat et al.": the title and DOI are Tueller, Ramirez, Cance et al. (2022), *Behav Res Methods* 55(8):4175 to 4199, doi:10.3758/s13428-022-02012-1 (no Pequignat author in PubMed for this topic). "De Bock et al.": the title and DOI are Natesan Batley and Hedges (2021), *Behav Res Methods* 53(4):1782 to 1798, doi:10.3758/s13428-020-01522-0. The annotations of 20 for these entries describe the fabricated attributions and are not to be carried	Appendix G corrections
GP-30	The unreported common-grid contrast (`36-fig4-contrast-reps100.png`)	writing, if `13-summarize-contrast.R` outputs suffice (inferred)	2 hours
GP-31	Paper 02 CO rerun with the corrected pre-exposure coding	computation	0.5 day plus server

ID	Gap	Status	Rough cost
GP-32	The OL+BDC Type I error under compound symmetry	analysis	1 day
GP-33	Paper 11 boundary cells, analytic super-additivity, MCSE columns (`CLAUDE.md`)	computation	1 day plus server
GP-34	Reconcile 37's anchored-change figures	done 2026-10-11 (CR-31)	
GP-35	`characterize_carryover()` against F1A; its grid	done 2026-10-11 (CR-43); code change proposed, not implemented	
GP-36	The published dropout form	done 2026-10-11 (CR-44)	
GP-37	Linter checks for $E($, $N($ and reserved letters	code	2 hours
GP-38	A single open-items register	writing	0.5 day
GP-39	F1 at 0.2 week in the 41-cell stress test with the full screens of 46 (only the job 14 extension evaluated F1h02)	computation, optional: the job 14 evidence covers size in 58 cells including the prognostic ones (quoted, 57 Section 8.2); maximin under the 46 rule at 0.2 week is not reported	0.5 day plus server

9. Migration plan

9.1 Phases

Table 23: Migration phases (computational gaps GP-15 to GP-22, GP-25, GP-31 to GP-33 and GP-39 run alongside, on the server)

Phase	Tasks	Depends on	Verification	Effort
0. Freeze and archive	create `docs/legacy/` and `docs/series/`; tag the repository (`whitepapers-pre-series`); move the archived files with `git mv`, unchanged except for a banner; move the files removed from the series; move the slide deck to `docs/archive/` with an unreviewed label; delete the two dated duplicate PDFs; write `docs/legacy/INDEX.md` (old number to new home)	DQ-1, DQ-2, DQ-4, DQ-9 (decided)	every file in Section 4 accounted for (a script lists `docs/` before and after); `git log --follow` works for a sample	0.5 day
1. Resolve conflicts and report finished runs	settle CR-31, CR-43, CR-44, CR-51 by inspection; tabulate job 15 and the covariance anchored pass; report jobs 21 and 22 when they finish; rerun the decision script on the refined process; GP-29 references	Phase 0; jobs 21, 22	each CR row has a stated answer and source; decision files `*-anchored.csv` produced	3 days plus job time

Phase	Tasks	Depends on	Verification	Effort
2. Skeleton and content moves	write the chapter template and registries (`chapters.csv`, `analyses.csv`); create every chapter with its outline; move content from the spines; write Appendix I; align driver defaults with the 0.2 week policy (inspect `t_half_a` defaults first)	Phase 1 for Parts IV and V; Phases 0 and 1 for Appendix I	every source section of Section 3 mapped to a chapter section (a checklist generated from the tables of Section 3)	8 days
3. New writing	GP-01 to GP-14, GP-23, GP-24, GP-26 to GP-28, GP-30, GP-38	Phase 2	each chapter passes the template checklist	10 days
4. Notation and convention pass	Section 7; linter with the new checks (GP-37); evidence labels; construct boxes; analysis IDs; prose per Appendix F	Phase 3	`perl tools/notation-lint.pl docs/series/*.md` clean; a grep for retired labels returns nothing outside quotations	3 days
5. Render, link check, retire	render every chapter through `tools/render.sh`; check every `docs/NN`, `series/chNN` and CM label reference resolves; update manuscripts, CM, NOTATION and `CLAUDE.md` (Sections 9.4 to 9.5); archive this plan	Phase 4	link checker over `docs/`, `analysis/report/`, `analysis/scripts/`; PDFs current	2 days

9.2 File layout and naming

Decided (DQ-2, 2026-10-11): a new folder `docs/series/` with identifiers that cannot be confused with the old numbers.

```
docs/
  series/
    fm0-reading-guide.md        fm1-executive-summary.md
    fm2-notation-conventions.md
    ch01-the-question.md ... ch28-patching-58b32a9.md
    appA-analysis-catalog.md ... appJ-software-pointer.md
    registry/chapters.csv  registry/analyses.csv
    registry/open-items.csv
  legacy/
    INDEX.md                    (old number -> new home)
    00-documentation-index.tex ... 59-whitepaper-refactoring-plan.md
  figures/                      (shared; new files chNN-figK-*.png)
  archive/                      (non-whitepaper history; the slide deck)
  references/  reviews/  tooling/
```

The alternative, keeping the old numbers and adding new documents as 60 and up, was rejected: numbers would not follow reading order, and the series would be interleaved with the documents it replaces. Renumbering old documents in place was rejected because 118 files outside `docs/` cite `docs/NN` (observed, grep: 71 scripts, 46 files under `analysis/report/`, and `CLAUDE.md`).

9.3 How old numbers keep resolving

Old numbers are permanent identifiers. `docs/NN` keeps meaning "legacy document NN"; the file lives at `docs/legacy/NN-*.md` (or `.tex`) and `docs/legacy/INDEX.md` maps each number, and each section where

a document was split, to its new home. The archived file carries one banner line ("Archived 2026-MM-DD; current treatment: `series/chNN` Section K; see `docs/legacy/INDEX.md`"). No redirect stubs are left in `docs/`; a stub per old file would put 58 files back into the folder the refactor is clearing. `fm0` prints the same map for readers.

9.4 Updating citations in manuscripts, CM and scripts

- **Scripts (71 files).** Their `docs/NN` comments are provenance and keep pointing to the legacy document, which is correct by Section 9.3. No edit is needed; a one-line note in the script header convention says how to resolve them.
- **Manuscripts (46 files under `analysis/report/`, including `bullets*.Rmd`, `report-slim.Rmd`, the `.tex` outputs and the `revisions/` files).** Citations to current content are changed to `series/chNN` with section; citations that refer to history (for example a referee report's quotation) keep `docs/NN`. The `.tex` files are regenerated by rendering, not edited. The edit list is produced by a grep and applied with the Edit tool (the repository path may resolve under cloud storage; in-place stream editors are not used, per the user's rule).
- **CM.** Its citations of docs 46 to 54 move to chapter references; it gains citations of `ch23`, `ch24`, `ch27` (GP-28).
- **NOTATION.** The "Used in" columns are rewritten from `docs/NN` to `series/chNN` in one revision with a dated entry; "Prose terminology" points to Appendix F.
- **P04.** Its 76 citations of `docs/58` (survey d) become `ch24`; its analysis labels take the `M-` prefix (DQ-5).

9.5 Keeping the project `CLAUDE.md` in sync

`CLAUDE.md` is project configuration, and this plan does not edit it. The thirteen stale or missing statements this plan found are listed below. Items 1 to 12 were corrected in `CLAUDE.md` on 2026-10-11 by the user (decided; not re-read here). Item 13 is open: it points to `docs/series/`, which does not yet exist, so `CLAUDE.md` carries for now a "Documentation map" paragraph pointing to this plan instead; the paragraph is replaced by the pointer of item 13 in Phase 5.

1. Corrected 2026-10-11. "Key architectural feature": "40-60% for Architecture B vs. 8-13% for Architecture A" was stale (CR-05).
2. Corrected 2026-10-11. "Architecture A", "Architecture B", "Architecture C" throughout: retired labels (NOTATION Part 3).
3. Corrected 2026-10-11. "expanded feasible range for `c.bm` from max 0.25 under CS to 0.49+ under AR(1)": stale (CR-04), as `docs/34` Section 6.2 had asked.
4. Corrected 2026-10-11. "Analysis model … Fixed formula: `Sx ~ bm + t + Dbc + bm:Dbc`" with a random intercept: no longer the recommended analysis (CR-10).
5. Corrected 2026-10-11. "`carryover_t1half` … (default 0.5-1.0 for the PTSD/prazosin context)": conflicted with the 0.2 week analysis policy and did not distinguish the data-generating half-life from $t^a_{1/2}$ (CR-37).
6. Corrected 2026-10-11. The repository layout listed `analysis/reports/` with three unnumbered slugs, and "Render a manuscript" named three slugs and called `rmarkdown::render()` directly, which the user's rule for `~/prj/res` forbids.
7. Corrected 2026-10-11. "`docs/figures/` PNG assets referenced by `docs/04-*.tex`": stale.
8. Corrected 2026-10-11. "See `docs/19-mean-moderation-implementation-notes.tex`": pointed to the unreviewed `.tex`; after Phase 0 the reference is Appendix I Section 5.
9. Corrected 2026-10-11. "Related external documents": the reorganization plan overstated as producing the current layout; `docs/23` described as current; `docs/plan-architecture-c.md`.
10. Corrected 2026-10-11. "Hendrickson et al. reported 2.5-5.8%": from `docs/03`, unverified against the publication.
11. Corrected 2026-10-11. British spellings.
12. Corrected 2026-10-11. The "Last updated" date.

13. Open. A short section pointing to `docs/series/fm0-reading-guide.md` and to `docs/legacy/INDEX.md`, added in Phase 5 in place of the interim "Documentation map" paragraph.

10. Decisions

All ten choices this plan put to the user were decided on 2026-10-11, in addition to the three decisions of Section 2.11. Each is recorded here with what it settles; the rest of the plan is written to them.

1. **DQ-1. Placement of the mathematics compendium.** Decided 2026-10-11: CM is the series' mathematical reference, Volume II, the single home of derivations, cited by label from each chapter's "Derivations" section (Section 2.3).
2. **DQ-2. Naming and layout.** Decided 2026-10-11: `docs/series/` with `fm`, `ch` and `app` identifiers, and `docs/legacy/` holding the old whitepapers under their old names, with an old-to-new index (Sections 9.2 to 9.3).
3. **DQ-3. Docs 53 and 55.** Decided 2026-10-11: both stay standalone notes for the original authors (`ch26`, `ch27`), and a series chapter, `ch28`, summarizes both, with the patches and the single power table; their duplicated tables are removed.
4. **DQ-4. The slide deck.** Decided 2026-10-11: `Architecting_Precision_Medicine.pdf` is moved out of the top level of `docs/` into `docs/archive/`, not deleted from the repository, and is labeled there as unreviewed (an AI-generated deck of 12.2 MB with no text layer, cited nowhere).
5. **DQ-5. Analysis IDs.** Decided 2026-10-11: the `M-` prefix marks main-effect analyses only (`M-F1`, `M-VF`, `M-RV`, …), in P04, `docs/58` and `ch24`, before paper 04's stored data exist; the interaction labels stay as stored; documents name the track at first use ("interaction analysis F1"). `M-` with a hyphen does not collide with the paper-local `10-M0` to `10-M3` labels of Section 2.7.
6. **DQ-6. σ_v^2 and τ^2.** Decided 2026-10-11: they are the same concept, the between-participant variance of the drug effect; τ^2 is written everywhere (in `ch11`, `ch14` and every carried passage of 38 and 51), and the identification is recorded in NOTATION.
7. **DQ-7. Name of the estimand.** Decided 2026-10-11: "biomarker-treatment interaction" everywhere, replacing "biomarker-by-drug" in the titles and text carried from 43 to 57.
8. **DQ-8. The covariance-study passes.** Decided 2026-10-11: `ch20` reports the anchored main pass; the follow-up and $D_{bc,it}$ passes stay cancelled, and `ch20` states this, together with the fact that 45 Sections 6 and 7 rest on the recursive passes and that the recommendation (F1 at 0.2 week, F1A0B) does not depend on them.
9. **DQ-9. Location of the archive.** Decided 2026-10-11: `docs/legacy/` inside the repository, so that `docs/NN` keeps resolving and history stays in git.
10. **DQ-10. Half-life grid.** Decided 2026-10-11: new runs use one grid for both tracks, $\{0, 0.2, 0.5, 1, 2\}$; the stored interaction runs keep $\{0, 0.25, 0.5, 1, 2\}$, stated wherever their results are quoted (in the construct box).

11. Evidence basis and limits of this plan

- **What rests on the surveys.** Most per-document findings (status, section contents, notation problems, citation counts) are taken from the four survey files and were not re-checked; they are labeled survey where cited.
- **What was spot-checked** against the source on 2026-10-11: `docs/03` Section 2.2 (lines 57 to 58); `docs/31` Section 18.1 (line 1837); `docs/37` header (lines 16 to 17) and Appendix A (line 1662); `docs/47` Section 1 (line 123); `docs/48` Section 4 (line 384); `docs/54` Section 6 (lines 876 to 880); `docs/55` (lines 444, 464, 685); `docs/56` Section 9 (lines 1004, 1022); `docs/57` Sections 1 (lines 136 to 234), 8.1 and 8.2, and line 1544; `docs/58` line 222; `RERUN-anchored.md` Sections 1, 2 and 7; the analyses and row counts of `summary-full-anchored.csv`, `summary-full-extra-anchored.csv`, `summary-full-extra2-anchored.csv` and `summary-alt250-null500-anchored.csv`.

- **Not checked.** The contents of `docs/references/` (including `pmsimstats-nof1-whitepaper.md`), `docs/archive/` and the 10 pages of the slide deck after the first; whether CM already holds 54 Section 5 in full; the code defaults for $t^a_{1/2}$; whether job 15 used the same cells and seeds as the stored run beyond the file layout; the page estimates, which are judgments.
- **Counts.** File counts are from `ls` of `docs/` (115 top-level files); the citing-file counts are from a grep for `docs/NN` over `analysis/report/`, `analysis/scripts/`, `R/` and `CLAUDE.md`.
- **Nothing was edited** besides this file, apart from pruning the earlier staged copies of this plan in `analysis/report/share/` and their `MANIFEST.md` rows; no simulation was run.

References

Giacomini R, Politis DN, White H (2013). A warp-speed method for conducting Monte Carlo experiments involving bootstrap estimators. *Econometric Theory* 29(3):567-589.

Morris TP, White IR, Crowther MJ (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine* 38(11):2074-2102.